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Paleomagnetism of Proterozoic mafic dikes from the Tobacco Root Mountains, southwest Montana

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Abstract

Paleomagnetic data from Proterozoic mafic dikes in southwestern Montana provides evidence for two distinct episodes of subparallel dike emplacement at ca. 1450 and 780 Ma. Published geochemical data from dikes in the southern Tobacco Root Mountains has identified three distinct compositional groups, termed groups A, B, and C. Geochronological data from the group A dikes yielded a Sm–Nd age of 1448 ± 49 Ma. Emplacement of these dikes is thought to reflect mafic magmatism associated with extension accompanying development of the adjacent Mesoproterozoic Belt Basin. Paleomagnetic results from these dikes and a group C dike yield antipodal magnetizations with a group-mean direction of $D = 225.0^\circ$, $I = 61.8^\circ$ ($k = 27.9$, $\alpha_{95} = 7.7^\circ$, $N = 14$ independent means/24 sites). The average paleomagnetic pole (8.7°N, 216.1°E, $A_{95} = 10.3^\circ$) is considered to be primary on the basis of positive baked contact tests and similarity to poles of ca. 1.45–1.4 Ga from intrusions elsewhere in North America, but is discordant with respect to poles from age equivalent sedimentary rocks of the Meosoproterozoic Belt Supergroup. $^{40}Ar/^{39}Ar$ dates from geochemical group B dikes are consistent with published U–Pb dates that demonstrate dike emplacement at 780 Ma as part of the regional Gunbarrel magmatic event. Hornblende concentrates from the group B dikes yield $^{40}Ar/^{39}Ar$ apparent ages of 778–772 Ma, whereas biotite from a baked contact zone yielded a plateau date of 788 Ma. Paleomagnetic results from the group B dikes yield a mean direction of $D = 301.5^\circ$, $I = -17.1^\circ$ ($k = 65.7$, $\alpha_{95} = 4.0^\circ$, $N = 12$ independent means/23 sites) with a paleomagnetic pole at 14.6°N, 127.0°E ($A_{95} = 3.2^\circ$). The combination of geochronologic data, results of a baked contact test, and spatial agreement of the paleomagnetic poles with poles of similar age elsewhere in North America indicates that this is also a primary magnetization associated with dike emplacement. Paleomagnetic data from some of the Tobacco Root Mountains dikes provide evidence that they were partially to completely remagnetized during latest Cretaceous to early Tertiary time, perhaps due to thermal affects associated with emplacement of the Late Cretaceous Tobacco Root Batholith. The overall agreement of paleomagnetic poles from the Proterozoic dikes with those of age equivalent rocks elsewhere in North America and agreement of the secondary magnetization with expected directions for the latest Cretaceous/early Tertiary indicate that the rocks of the Tobacco Root Mountains have not experienced significant tilting or vertical axis rotation since the Mesoproterozoic. The new paleomagnetic poles from this study thus provide key data for refining Meso- and Neoproterozoic parts of the North American APW path.
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1. Introduction

Proterozoic mafic dikes are a common feature of several Laramide-age (latest Cretceous–early Tertiary) basement-cored uplifts of southwestern Montana (Fig. 1A and B). Most dikes strike west or northwest and cut Archean crystalline basement. Many dikes parallel or are subparallel to basement faults of presumed Precambrian ancestry. Dike emplacement and faults are thought to have accompanied extension that formed the adjacent Mesoproterozoic Belt Basin (Harrison et al., 1974; Wooden et al., 1978; Schmidt and Garihan, 1986; Harlan et al., 2005). Based on geochemical data from dikes in the Ruby Range and southern Tobacco Root Mountains, Wooden et al. (1978) identified three distinct geochemical groups, with apparent ages of ca. 1450 Ma and 1120–1130 Ma. These ages are similar to dates from mafic sills that intrude sedimentary rocks of the Belt Supergroup (Hunt, 1962; Zartman et al., 1982; Höy, 1989; Anderson and Davis, 1995; Sears

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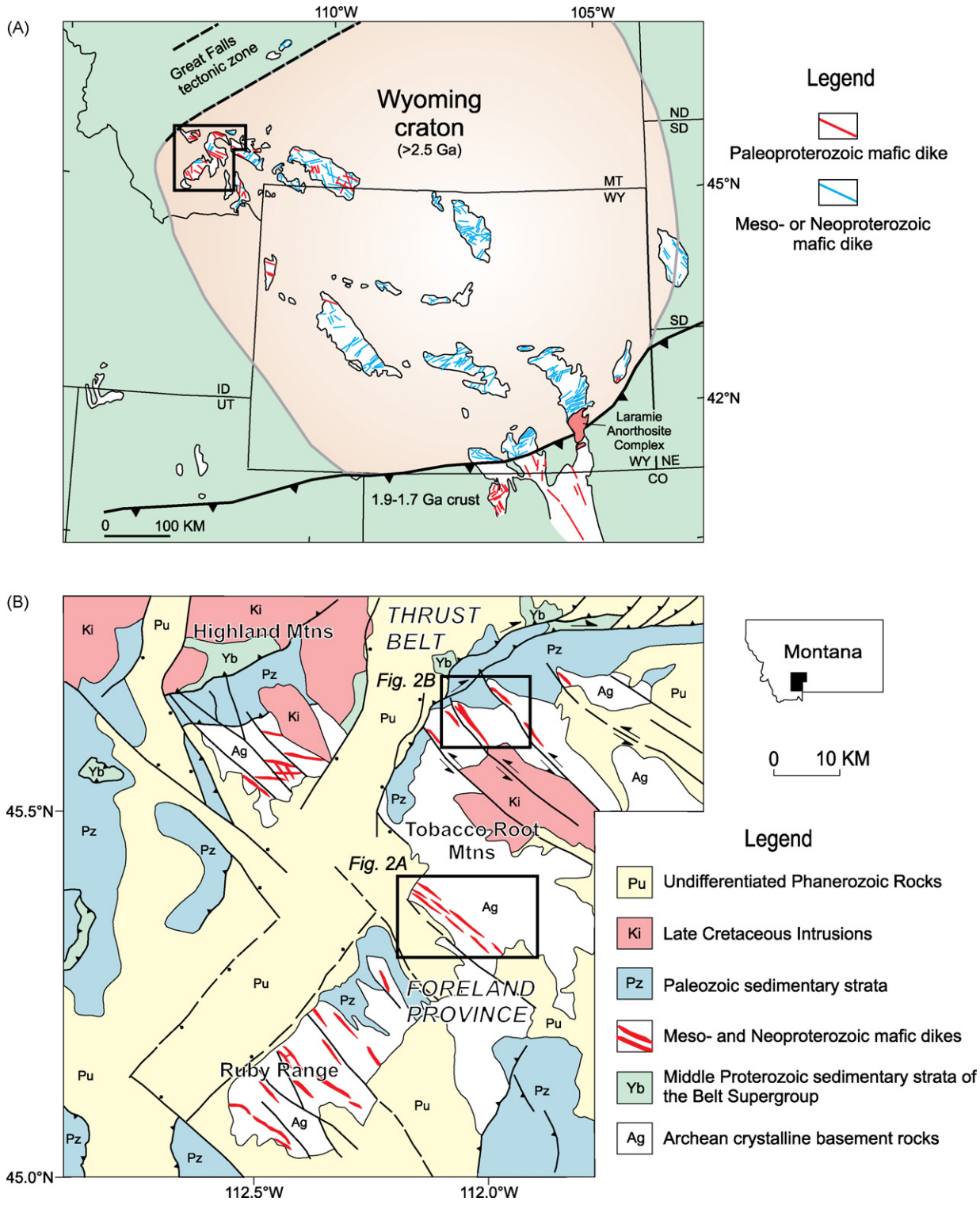


Fig. 1. (A) Map showing the approximate limits of the Wyoming craton and the general distribution of Paleoproterozoic and Meso- and Neoproterozoic dikes exposed in the basement-cored uplifts. Inset shows the location of (B). Figure modified from Snyder et al. (1988). (B) Tectonic map of southwestern Montana showing the location of the Tobacco Root Mountains, the Cordilleran fold and thrust belt, the Rocky Mountains foreland province, and the distribution of Meso- and Neoproterozoic diabase dikes. The insets show the locations of mafic dikes sampled in this study and locations of Fig. 2A and B. Figure modified from Schmidt and Garihan (1986) and Schmidt et al. (1988).

et al., 1998) and to metamorphic dates from those strata (Obradovich and Peterman, 1968). The two distinct time periods were considered to reflect mafic magmatism accompanying distinct episodes of tectonism and/or sedimentation within the Belt Basin; however, recent high-precision geochronology has demonstrated that most of the Belt Supergroup was deposited between 1470 and 1400 Ma (Evans et al., 2000).

We report new paleomagnetic results from Proterozoic mafic dikes exposed in the Tobacco Root Mountains of southwest Montana. Our goal in undertaking this study was to better understand the age, distribution, and tectonic importance of Proterozoic mafic dikes emplaced along the northwest margin of the Wyoming craton, and to contribute temporally well-constrained paleomagnetic poles to the North American apparent polar wander (APW) path. Because the ca. 1.4 and 1.1 Ga segments of the North American APW path are relatively well defined, and assuming that post-emplacement deformation has been minimal, these data could be used to confirm whether multiple episodes of dike intrusion occurred in southwest Montana. If primary magnetizations could not be isolated because of subsequent remagnetization(s), the paleomagnetic data could help resolve the age and nature of post-emplacement events affecting the dike isotopic systems. Our paleomagnetic data yield three distinct magnetizations. Two of these are primary, having been acquired during dike emplacement during Proterozoic time, whereas the third is a product of latest Cretaceous remagnetization. Comparison of the primary dike paleomagnetic poles with those from Proterozoic rocks elsewhere in North America indicates that dikes of two of Wooden et al.'s (1978) geochemical groups were emplaced at ca. 1450 Ma, with a third distinct group emplaced at 780 Ma. The paleomagnetic and $^{40}Ar/^{39}Ar$ data reported here are consistent with Sm–Nd and U–Pb results from the southern Tobacco Root dikes, which provided evidence for dike intrusion at 1448 and 780 Ma (Harlan et al., 2005). New $^{40}Ar/^{39}Ar$ dates from hornblende from mafic dikes and a biotite from a contact gneiss sample reported here provide further evidence in support of Neoproterozoic magmatism in the southern Tobacco Root Mountains. The ca. 1448 Ma dikes are related to extension in the Belt Basin, whereas the younger group B dikes are part of a regional magmatic episode termed the Gunbarrel mafic magmatic event that extended from northwestern Wyoming to the Arctic Ocean and was a precursor to Neoproterozoic breakup of the Rodinian supercontinent (Harlan et al., 2003b). Paleomagnetic poles from this event are key reference poles that may be used to assess the validity of competing continental reconstructions for Rodinia (Park et al., 1995b; Harlan et al., 2003b; Bleeker, 2004). Although summary poles utilizing paleomagnetic data from the Gunbarrel dikes have been cited by numerous authors, the details of paleomagnetic results from the southern Tobacco Root dikes have not been published.

2. Geological setting

The Tobacco Root Mountains are a broadly domal, basement-cored uplift located at the intersection of two Laramide (Late Cretaceous–early Tertiary) tectonic provinces: (1) the Rocky Mountain foreland, characterized by basement-involved deformation along low- to moderate-angle reverse faults; and (2) thin-skinned thrusts and folds of the Helena salient of the Cordilleran thrust belt, within which deformation is limited largely to sedimentary strata of Mesoproterozoic to Mesozoic age (Fig. 1B). Superimposed on structures of both tectonic provinces are normal faults developed during late Cenozoic to Recent extension that produced the present-day basins and ranges of southwestern Montana (Reynolds, 1979; Schmidt and Garihan, 1986; Kellogg et al., 1995).

The Tobacco Root Mountains expose one of the northernmost parts of Archean crystalline basement in the Wyoming craton (Fig. 1A). The basement rocks are comprised of rocks of the Montana Metasedimentary Province, a terrane characterized by metasedimentary rocks, including marble, quartizite, iron formation, and pelitic schist (Mogk et al., 1992, 2004). In the Tobacco Root Mountains, the basement rocks are dominated by three distinct lithologic assemblages (Vitaliano et al., 1979; Mogk et al., 2004; Burger et al., 2004): (1) the Indian Creek Metamorphic Suite; (2) the Pony-Middle Mountain Metamorphic Suite; and (3) the Sphuler Peak Metamorphic Suite. The Indian Creek and Pony-Middle Mountain suites are dominated by quartzofeldspathic gneiss and variable amounts of metamorphosed supracrustal rocks. The Sphuler Peak Metamorphic Suite is dominated by amphibolite, with lesser amounts of quartzite and aluminous schist. The Indian Creek and Pony-Middle Mountain metamorphic suites are interpreted to be part of a platform-type sedimentary sequence deposited on a late Archean continental margin (Mogk and Henry, 1988; Wooden et al., 1988; Mogk et al., 2004), whereas the Sphuler Peak Metamorphic Suite is thought to have originated as oceanic crust (Burger et al., 2004). Gneisses from the central part of the range have protolith crystallization ages of 3.2–3.4 Ga and record evidence for metamorphism at ca. 2.8 Ga (Mueller et al., 2004). Pb–Pb monozite and zircon age determinations from record evidence of high-grade metamorphism and/or magmatism at ca. 2.4 Ga, but no rocks of this age have been identified in the Tobacco Root Mountains (Cheney et al., 2004a; Mueller et al., 2004). Rocks of the Indian Peak and Pony-Middle Mountain metamorphic suites are cut by amphibolite dikes and sills termed Proterozoic Metamorphic Mafic Dikes and Sills (Brady et al., 2004b); one of these intrusions yielded a U–Pb zircon crystallization age of 2.060 ± 0.006 Ga (Mueller et al., 2004). The amphibolites are interpreted to represent tholeittic dikes and sills intruded into cool crust, perhaps associated with continental rifting of the Wyoming craton (Brady et al., 2004a). The 2.06 Ga age for these dikes is similar to several ca. 2.1–2.0 Ga U–Pb dates reported from NE-striking dikes along the south and southeastern margins of the Wyoming craton that have also been attributed to continental extension, rifting, and development of a passive continental margin at ca. 2.2–2.1 Ga. It has been hypothesized that this extensional event records break up and dismemberment of the Wyoming and Superior cratons at this time prior to their reassembly at ca. 1.8 Ga (Karlstrom et al., 1981; Premo and Van Schmus, 1989; Cox et al., 2000; Roscoe and Card, 1993; Chamberlain et al., 2003; Harlan et al., 2003a; Bowers and Chamberlain, 2006; Duebendorfer et al., 2006).

Rocks of the Indian Creek Metamorphic Suite, the Pony-Middle Mountain Metamorphic Suite, the Sphuler Peak Metamorphic Suite and the mafic rocks of the Proterozoic Metamorphic Mafic Dikes and Sills were tectonically juxtaposed and regionally metamorphosed to at least upper amphibolite facies, at ca. 1.77 Ga, during a widespread tectonothermal event that some workers have termed the "Big Sky Orogeny" (Brady et al., 2004a,b; Cheney et al., 2004a,b; Harms et al., 2004). This

deformational event is thought to represent deep burial during a continental-scale collision, perhaps due to collision of the Hearne Province or other terrane to the north and west with the Wyoming craton. Collision may have been followed by isothermal decompression, probably due to rapid tectonic denudation late in the orogenic cycle (Cheney et al., 2004a), as suggested by mineral dates from rocks in southwest Montana from several geochronologic systems. U–Pb zircon, $^{207}Pb/^{206}Pb$ monazite, K–Ar, and $^{40}Ar/^{39}Ar$ yield similar apparent ages ranging from 1.82 to 1.70 Ga (Giletti, 1966, 1971; Erslev and Sutter, 1990; Vargo, 1990; Harlan et al., 1996; Roberts et al., 2002; Brady et al., 2004a; Cheney et al., 2004b; Mueller et al., 2004). Alternatively, Mueller et al. (2004) propose that the 1.77 Ga tectonothermal event may be related to post-collision mafic underplating, extension, and rapid exhumation of the entire region.

The Archean basement rocks in the Tobacco Root Mountains are cut by unmetamorphosed Proterozoic diabase dikes that strike N70°W, are near vertical, and cut the northeast-striking metamorphic foliation of the basement rocks (Koehler, 1973; Vitaliano et al., 1979) (Fig. 2A). Individual dikes range from 0.3 to 110 m in thickness, and are up to 10-km long. Although many dikes are shown as continuous features in geologic mapping by Vitaliano et al. (1979) (Fig. 2A), our observations indicate that many are discontinuous. In the northern Tobacco Root Mountains, the dikes strike N30°W to N50°W and are exposed subparallel to, or along, northwest-striking Laramide reverse faults and occupy the axial regions of several foreland anticlines (Fig. 2B). One of these, the Mammoth dike, is 250–300 m thick and is locally highly differentiated (Koehler, 1973).

The Precambrian rocks of the Tobacco Root Mountains were overlain by sedimentary rocks during the Paleozoic and Mesozoic, but these rocks were stripped from the range during Laramide deformation and now crop out only in the northern and northwestern parts of the range. The core of the range consists of the granodioritic Tobacco Root Batholith of Late Cretaceous age, which has yielded K–Ar dates of 77–72 Ma (Vitaliano et al., 1980). Numerous Late Cretaceous or early Tertiary satellitic sills, dikes, and plugs are also present in the northern and southern Tobacco Root Mountains. Eocene and Oligocene volcanic rocks form a broad volcanic plateau north of Virginia City, in the southeast Tobacco Root Mountains (Fig. 2A).

The west- and northwest-striking dikes of southwestern Montana are a pervasive tectonic element of these basement-cored uplifts and their origin and tectonic importance has long been the subject of speculation. Based on Rb–Sr dates, the dikes were considered to reflect mafic magmatism accompanying development of the nearby Mesoproterozoic Belt Basin (Wooden et al., 1978; Schmidt and Hendrix, 1981; Schmidt and Garihan, 1986). The dikes are found in, or parallel to subparallel to, faults interpreted to have been reactivated as basement-involved reverse or thrust structures in the latest Cretaceous. Schmidt and Garihan (1986) proposed that the dikes were intruded parallel to or along normal faults formed during northeast- and southwest-directed basement extension accompanying development of the Helena Embayment of the Mesoproterozoic Belt Basin, exposed to the west and northwest. Many of the Mesoproterozoic normal faults were reactivated during Laramide shortening as reverse faults and/or as normal faults during late Cenozoic extension. The Rb–Sr dates of 1450 and 1130 Ma were interpreted to record distinct episodes of tectonism and sedimentation in the Belt Basin and both dates are similar to isotopic dates reported from mafic sills and sedimentary rocks elsewhere in the basin (Wooden et al., 1978).

In the southern Tobacco Root Mountains, Wooden et al. (1978) identified three groups of dikes based on major and trace element geochemistry, which they termed A, B, and C. Group A dikes are fresh to somewhat altered and are fine- to medium-grained (≤1 to 2 mm average grain size) diabase. They consist of clinopyroxene (augite) and plagioclase, with minor biotite and amphibole. Mafic minerals are slightly altered to chlorite, and plagioclase is saussuritized. These dikes are ophitic and subophitic to intergranular; porphyritic and glomeroporphyritic textures are sometimes present. Group A dikes are thin, with thicknesses ranging from 2 to 7 m. The dikes are chemically undifferentiated, display moderate iron-enrichment, and are among the most primitive of the dikes in the southern Tobacco Root Mountains (Wooden et al., 1978). The southern Tobacco Root dikes are petrographically and chemically similar to dikes from the nearby Ruby Range to the southwest, and represent a common magma emplaced at the same time (Wooden et al., 1978).

Group B dikes are composed of clinopyroxene (augite) and plagioclase, but also contain hornblende and biotite. A striking feature of these dikes is the presence of graphic intergrowths of quartz, plagioclase, and alkali-feldspar. Magnetite, ilmenite, sulfide minerals (chalcopyrite), and apatite are common accessory phases. Group B dikes are commonly altered: plagioclase is replaced by clay minerals and white mica, and hornblende, biotite, and pyroxene are altered to chlorite and blue-green fibrous amphibole. This alteration is deuteric in origin, and occurred during protracted cooling and hydrothermal circulation following emplacement. The dikes have well-defined chilled margins and have medium- to coarse-grained (2–5 mm) interiors where they exhibit ophitic to gabbroic textures. In the southern Tobacco Root Mountains, their coarse grain size, abundance of hydrous minerals, graphic intergrowths, and thickness (10–110 m) distinguish them from either group A or C dikes. The group B dikes are iron-enriched and are quartz normative (Wooden et al., 1978).

Group C dikes are represented by samples from only two dikes (Wooden et al., 1978). These dikes are fine to medium grained, have intergranular textures, and consist of clinopyroxene (augite–pigeonite) and plagioclase, with minor hornblende, biotite, and oxides. Plagioclase is moderately altered and some pyroxenes are mantled by secondary amphibole or chlorite. Group C dikes are characterized by low iron and moderately high alkali and silica contents (Wooden et al., 1978).

Rb–Sr results from group A dikes in the southern Tobacco Root Mountains gave a limited spread of Rb–Sr ratios and were not amenable to an age calculation (Wooden, 1975). Rb–Sr ratios from similar dikes in the Ruby Range, have a greater spread in $^{87}Rb/^{86}Sr$ versus $^{87}Sr/^{86}Sr$ ratios, providing an isochron age of 1455 ± 125 Ma with an initial $^{87}Sr/^{86}Sr$ ratio of 0.7019 ± 0.0008

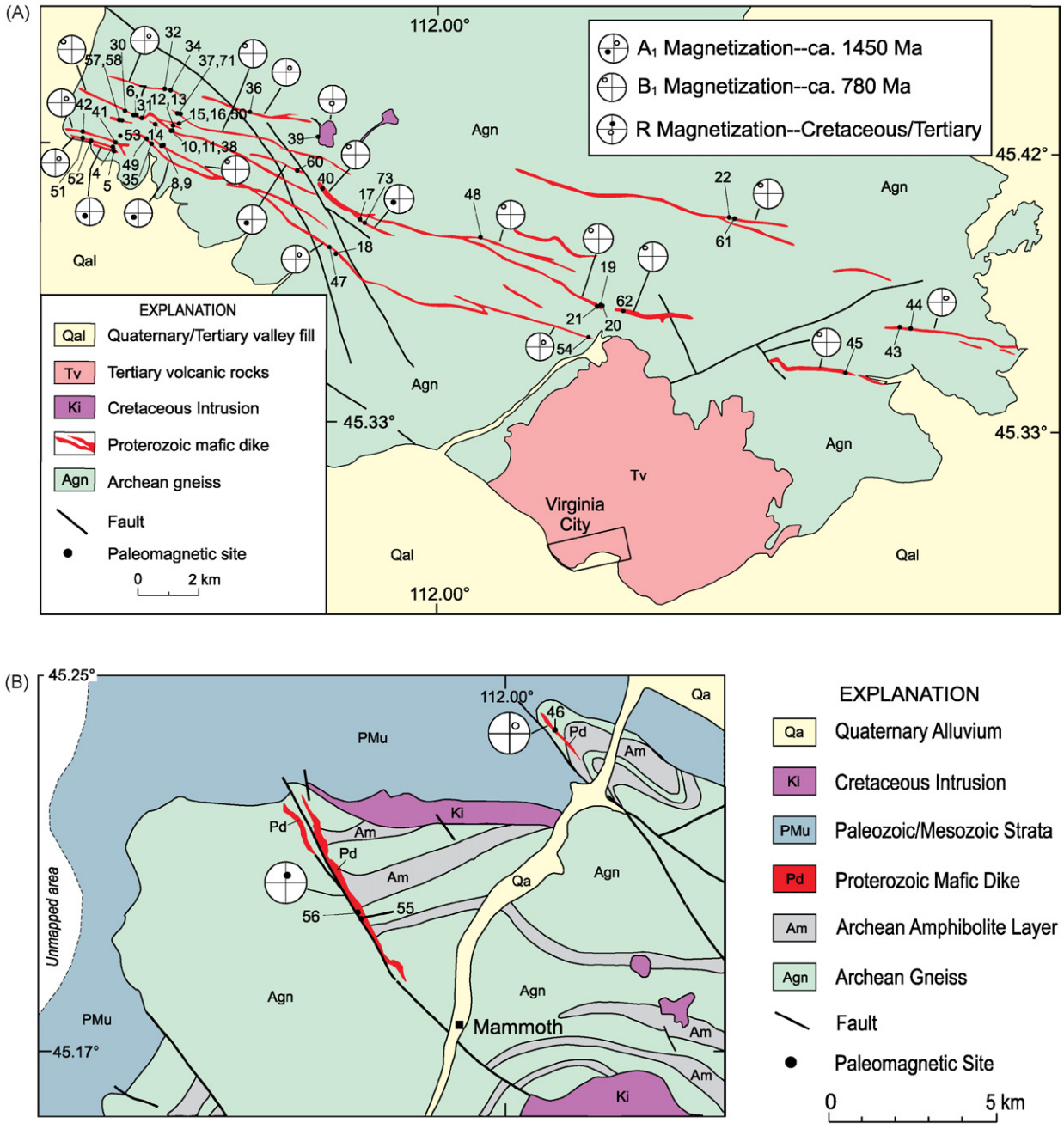


Fig. 2. (A) Generalized geological map of the southern Tobacco Root Mountains showing the distribution of west–northwest-striking diabase dikes and the location of paleomagnetic sites sampled in this study. Also shown are idealized equal area stereograms which show the overall paleomagnetic direction obtained for individual dikes or dike segments. (B) Geologic map of diabase dikes exposed in the axial regions of twofolds two Laramide folds in the northern Tobacco Root Mountains and the location of paleomagnetic sites. Stereograms as in Fig. 1. Both figures modified from Vitaliano et al. (1979).

(Wooden et al., 1978). Most Rb–Sr data from the Tobacco Root group A dikes scatter about this isochron. Wooden et al. (1978) argued that the age obtained from the Ruby Range dikes best represents the age of emplacement of the group A dikes in the southern Tobacco Root Mountains. Group B Rb–Sr data are scattered; hence, their geochronologic interpretation is more complicated than the group A dikes. Data from three localities appear to define a linear array that Wooden et al. (1978) used to calculate an age of 1120 ± 185 Ma, with an initial $^{87}Sr/^{86}Sr$ ratio of 0.7025 ± 0.0022. Other, more highly altered samples formed parallel isochrons with higher initial ratios. The petrographically freshest group C samples gave an apparent Rb–Sr age of 1130 ± 130 Ma, with an $^{87}Sr/^{86}Sr$ initial ratio of 0.709 ± 0.002. Although this age is statistically indistinguishable from that of the group B dikes, their compositions and initial ratios indicate that they are not comagmatic (Wooden et al., 1978).

Recent geochronologic work by Harlan et al. (2005) refined the emplacement history of the Tobacco Root Dikes. Sm–Nd mineral dating of a group A dike yielded an apparent age of 1448 ± 49 Ma, which is considered to be the best estimate of the age of emplacement of these dikes. Because group A and group C dikes yield similar remanent magnetizations (as we

report below), Harlan et al. (2005) suggested that dikes of both geochemical groups were emplaced at about the same time, and this age is in good agreement with U–Pb dates from sills in Belt Supergroup strata exposed to the north and west of the Tobacco Root Mountains (Zartman et al., 1982; Höy, 1989; Hunt, 1962; Anderson and Davis, 1995; Sears et al., 1998) and from dikes exposed in the basement of foreland uplifts elsewhere in Wyoming (Chamberlain and Frost, 1995). U–Pb dating of zircon and baddeleyite from the more differentiated group B dikes yielded an apparent age of 787 ± 11 Ma, although the zircons used in the analysis were discordant. An identical, but analytically more precise, upper intercept age of 782.4 ± 4.9 Ma was determined using four baddeleyite analyses. Because baddeleyite is a primary igneous phase in mafic dikes and thus is unaffected by inheritance, the upper intercept age was considered to represent the best estimate of the age of emplacement of the group B dikes (Harlan et al., 2005).

To supplement the geochronological date reported by Harlan et al. (2005), we report additional data consistent with the Neoproterozoic age for the group B dikes described above. These provide evidence for a lack of significant tectonothermal disturbance of the dike and basement rocks in the southern Tobacco Root Mountains. Unfortunately, samples appropriate for $^{40}Ar/^{39}Ar$ dating could not be obtained from any Group A dikes. $^{40}Ar/^{39}Ar$ dates from hornblende from the group B dikes and biotite from basement gneiss in the thermal contact zone of a 100-m thick group B dike yield additional Neoproterozoic age estimates (Fig. 3). Analytical data and methods are presented in Table 1. Four hornblende concentrates from hand samples from paleomagnetic sites in three group B dikes (TR6, TR19, TR21, and TR62) yield similar apparent age spectra (Fig. 3A and B). The age spectra are complicated, yielding somewhat U-shaped spectra some excess ^{40}Ar in high- and low-temperature steps, but which also yield relatively flat and consistent ages for intermediate temperature steps (Table 1). Two of these samples yield identical plateau dates of 775 ± 5 Ma (2σ). The other two fail to satisfy the strict plateau criteria of Fleck et al. (1977), but give weighted mean apparent ages of 778 ± 3 and 772 ± 5 Ma for the intermediate temperature steps. We consider both the plateau and weighted mean dates to represent reasonable estimates for the emplacement age of these dikes. These dates are consistent with the U–Pb baddeleyite data from the dikes, especially considering incorporation of decay constant errors for both isotopic systems. Biotite obtained from the contact zone of paleomagnetic site TR50 (Fig. 3C), described below, also yields a plateau date of 787 ± 4 Ma, and suggests that the $^{40}Ar/^{39}Ar$ system of the adjacent basement rocks was almost completely reset by intrusion of the dike, although incorporation of excess ^{40}Ar during cooling may explain why it gives an apparent age somewhat older than the corresponding dikes. Elsewhere in the Tobacco Root Mountains, hornblende and biotite $^{40}Ar/^{39}Ar$ dates of 1.7 Ga (Brady et al., 2004a) have been reported from basement rocks indicating that they have not been thermally disturbed by younger tectonothermal processes, with the possible exception of rocks adjacent to Late Cretaceous intrusions, since cooling through hornblende and biotite closure temperatures (500 and 300 °C, respectively) following the peak of thermal effects of the Big Sky Orogeny.

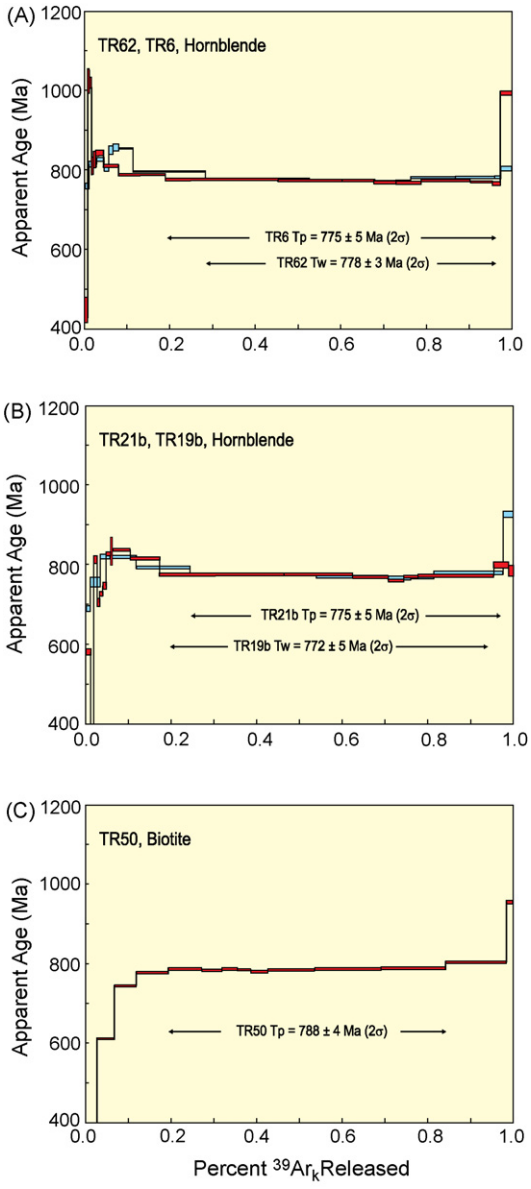


Fig. 3. $^{40}Ar/^{39}Ar$ age spectra for hornblende and biotite samples from the southern Tobacco Root Mountains. (A and B) Hornblende concentrates from group B dikes. The analytical error in the apparent age for each temperature step in the age spectrum is represented by the height of the shaded rectangle; the proportion of $^{39}Ar_K$ released in the incremental heating experiment is represented by the length of the shaded rectangle. Tp = plateau age calculated as described in the text; Tw = weighted mean age for samples that do not yield a plateau, but yield relatively consistent apparent ages. (C) Biotite from a paleomagnetic core within 10 cm of a group B dike in the thermal contact zone.

3. Paleomagnetic sampling and analysis

Paleomagnetic samples were collected from 46 sites in dikes from the southern Tobacco Root Mountains (Fig. 2A), and from three sites in the northern part of the range (Fig. 2B). Our sites are concentrated in the southwest corner of the range, where excellent dike exposures are found in several drainages

Table 1
$^{40}Ar/^{39}Ar$ incremental heating data for samples from the Tobacco Root Mountains, Montana

Temperature (°C)	$^{40}Ar_R$	$^{39}Ar_K$	$^{40}Ar_R/^{39}Ar_K$	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent age (Ma at 1σ)
TR50: Group B diabase dike; biotite from gneiss contact zone; 14.3 mg, measured $^{40}Ar/^{39}Ar_a = 299.5$; *J*-value = 0.02482 ± 0.25%							
500	1.8139	0.17722	6.680	11.99	83.2	3.0	278 ± 1
600	3.8654	0.23871	16.19	14.64	95.5	4.0	613 ± 3
650	6.2981	0.30670	20.54	18.81	94.0	5.1	747 ± 3
700	9.8274	0.45294	21.70	22.02	98.4	7.5	781 ± 3
750	10.283	0.46786	21.98	20.06	99.0	7.8	789 ± 3
775	6.1898	0.28274	21.89	18.53	98.4	4.7	787 ± 3
800	4.7176	0.21465	21.98	18.64	98.7	3.6	789 ± 3
825	4.1048	0.18722	21.93	18.71	98.3	3.1	788 ± 3
850	5.2503	0.24127	21.76	18.92	98.2	4.0	783 ± 3
900	14.316	0.65276	21.93	26.78	99.3	10.9	788 ± 3
950	20.517	0.93261	22.00	36.07	99.6	15.5	790 ± 3
1000	19.998	0.90659	22.06	39.45	99.3	15.1	792 ± 3
1050	19.169	0.84827	22.60	35.16	99.5	14.1	807 ± 3
1200	2.5600	0.09107	28.11	5.56	97.2	1.5	959 ± 4
Total gas			21.38				772 ± 3
TR62: Group B diabase dike; hornblende; 37.0 mg; measured $^{40}Ar/^{39}Ar_a = 298.9$; *J*-value = 0.007810 ± 0.25%							
850	0.11201	0.00062	179.3	0.24	62.9	0.1	1499 ± 50
950	0.29948	0.00412	72.65	0.38	71.7	0.9	761 ± 7
1000	0.61419	0.00776	79.19	0.50	62.7	1.7	816 ± 7
1100	0.62238	0.00772	80.62	0.31	90.8	1.7	828 ± 5
1133	0.36530	0.00469	77.84	0.19	86.7	1.0	805 ± 7
1167	0.37852	0.00453	83.51	0.16	92.1	1.0	851 ± 10
1200	0.50651	0.00600	84.42	0.16	92.3	1.3	859 ± 8
1233	1.2324	0.01466	84.04	0.16	96.2	3.2	856 ± 3
1267	5.9604	0.07745	76.96	0.18	98.5	17.0	797 ± 2
1300	8.3087	0.11109	74.80	0.18	99.3	24.4	779 ± 1
1333	5.2222	0.07012	74.48	0.19	99.4	15.4	777 ± 1
1367	1.6884	0.02281	74.01	0.19	98.0	5.0	773 ± 2
1400	1.1712	0.01586	73.82	0.16	96.8	3.5	771 ± 4
1433	3.5703	0.04750	75.17	0.17	98.4	10.5	782 ± 2
1467	3.1405	0.04173	75.25	0.15	97.9	9.2	783 ± 3
1500	0.52795	0.00701	75.29	0.14	95.1	1.5	783 ± 4
1650	0.84306	0.01083	77.82	0.09	93.6	2.4	805 ± 6
Total gas			76.04				790 ± 2
TR6: Group B diabase dike; hornblende; 53.2 mg; measured $^{40}Ar/^{39}Ar_a = 298.9$; *J*-value = 0.024740 ± 0.25%							
500	0.10486	0.00928	11.30	0.53	41.5	0.7	447 ± 32
600	0.10486	0.00516	30.84	0.52	83.5	0.4	1028 ± 31
700	0.15899	0.00685	30.67	0.28	44.7	0.5	1023 ± 14
750	0.21006	0.00509	22.86	0.16	74.6	0.4	791 ± 22
800	0.11645	0.00792	23.45	0.10	80.4	0.6	829 ± 21
850	0.18559	0.02483	24.06	0.14	94.4	1.8	847 ± 7
875	0.59722	0.04775	22.89	0.17	96.3	3.5	813 ± 4
900	1.0928	0.14935	22.12	0.17	98.5	11.0	791 ± 3
925	3.3042	0.35977	21.71	0.18	99.0	26.5	779 ± 3
950	7.8088	0.20395	21.58	0.19	99.1	15.0	776 ± 3
975	4.4006	0.0999	21.57	0.21	99.2	7.4	775 ± 3
1000	2.1544	0.07186	21.46	0.19	99.6	5.3	772 ± 4
1025	1.5421	0.07884	21.41	0.19	99.1	5.8	771 ± 4
1050	1.6879	0.15667	21.58	0.20	99.7	11.5	776 ± 3
1100	3.3589	0.07136	21.51	0.19	98.0	5.3	773 ± 3
1150	1.5350	0.02472	21.34	0.19	92.3	1.8	768 ± 5
1250	0.52760	0.03586	22.30	0.20	92.6	2.6	796 ± 5
Total gas	0.79982		21.78				781 ± 4
TR19b: Group B diabase dike; hornblende; 58.3 mg; measured $^{40}Ar/^{39}Ar_a = 298.9$; *J*-value = 0.015102 ± 0.25%							
500	0.28513	0.01133	25.177	0.59	58.3	1.1	584 ± 8
600	0.9606	0.00703	13.672	1.22	73.7	0.7	340 ± 20
700	0.28262	0.00753	37.551	0.57	59.1	0.8	814 ± 10
750	0.20046	0.00635	31.574	0.49	53.6	0.6	707 ± 11

Table 1 (*Continued*)

Temperature (°C)	$^{40}Ar_R$	$^{39}Ar_K$	$^{40}Ar_R/^{39}Ar_K$	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent age (Ma at 1σ)
800	0.23356	0.00713	32.767	0.34	83.4	0.7	729 ± 6
850	0.27792	0.00821	33.857	0.16	83.5	0.8	749 ± 9
900	0.43163	0.01124	38.398	0.09	89.1	1.1	829 ± 6
925	0.16019	0.00413	38.799	0.09	84.6	0.4	836 ± 35
950	1.5914	0.04082	38.983	0.12	94.9	4.1	839 ± 3
975	2.5831	0.06847	37.727	0.17	97.6	6.9	817 ± 4
1000	10.216	0.28827	35.440	0.19	98.9	29.1	777 ± 4
1025	5.6267	0.15874	35.446	0.20	99.2	16.0	777 ± 3
1050	2.9374	0.08375	35.072	0.21	99.3	8.4	771 ± 3
1075	1.2273	0.03548	34.594	0.18	99.6	3.6	762 ± 3
1100	1.1525	0.03279	35.147	0.16	99.9	3.3	772 ± 3
1150	6.1182	0.17525	35.276	0.15	98.5	17.7	774 ± 4
1200	1.2849	0.03490	36.816	0.08	96.6	3.5	801 ± 8
1300	0.39212	0.01089	35.997	0.10	95.9	1.1	787 ± 14
Total gas			35.434				777 ± 4
TR21b: Group B diabase dike; hornblende; 38.1 mg; measured $^{40}Ar/^{39}Ar_a = 298.9$; *J*-value = 0.015185 ± 0.25%							
700	0.21747	0.00700	30.656	0.42	58.7	1.2	693 ± 6
800	0.43590	0.01272	34.259	0.22	54.6	2.3	759 ± 12
900	0.47778	0.01261	37.899	0.10	82.6	2.2	824 ± 7
950	1.3806	0.03482	39.649	0.13	96.3	6.2	854 ± 3
975	2.5559	0.07034	36.335	0.18	98.1	12.5	796 ± 3
1000	5.8689	0.16606	35.342	0.19	99.0	29.5	779 ± 3
1025	3.2967	0.09418	35.004	0.20	99.3	16.7	773 ± 3
1050	1.0412	0.02990	34.827	0.20	99.7	5.3	770 ± 5
1100	1.0602	0.03038	34.896	0.16	98.2	5.4	771 ± 4
1150	3.2375	0.09104	35.562	0.14	98.3	16.2	783 ± 4
1250	0.60231	0.01361	44.245	0.03	96.3	2.4	931 ± 8
Total gas			35.850				788 ± 4

Note: $^{40}Ar/^{39}Ar_a$ is measured atmospheric argon used for mass-discrimination at the time of analysis; $^{40}Ar_R$ is radiogenic ^{40}Ar in volts signal; $^{39}Ar_K$ is potassium-derived ^{39}Ar in volts signal; $^{40}Ar_R/^{39}Ar_K$ is the ratio of $^{40}Ar_R$ to $^{39}Ar_K$ after correction for mass-discrimination and interfering isotopes; $^{39}Ar/^{37}Ar$ = ratio of $^{39}Ar_K$ to $^{37}Ar_{Ca}$ (this value can be converted to the approximate K/Ca by multiplying by 0.5); $\%^{40}Ar_K$ and $\%^{39}Ar$ are the percent of radiogenic ^{40}Ar and percent of total ^{39}Ar released in each temperature step. Temperature steps in italics are those used in the calculation of the plateau or weighted-mean age. Conversion of volts signal to moles Ar can be made using a conversion factor of 2.078×10^{-12} moles argon per volt of signal. The hornblendes and biotite used in this study were crushed, sieved, and mineral separates concentrated using standard density (heavy liquid), and magnetic techniques. Biotite sample TR50 was obtained from paleomagnetic drill core samples used in a baked contact test and were collected within 20 cm of the dike contact. The samples were subjected to AF demagnetization prior to preparation of the mineral separations. Optically pure (>99.9%) mineral separates were produced by hand-picking. Mineral separates and neutron flux monitors were wrapped in aluminum foil packages, placed in quartz vials, and irradiated in the U.S. Geological Survey TRIGA reactor. Variations in neutron flux were monitored by placing 6–10 standards along the length of each vial; the geometry of the package was such that each sample was adjacent to at least one standard. The flux monitor used in this study was MMhb-1, which has an assigned K–Ar age of 523.1 ± 2.6 Ma (Renne et al., 1998). Corrections for interfering isotopes were made by measuring the argon isotopes of K_2SO_4 and CaF_2 irradiated in each package. Each sample was heated in a double-vacuum resistance furnace for 20 min in a series of 10–20 steps to a maximum temperature of 1450 °C. Five isotopes of argon (^{40}Ar, ^{39}Ar, ^{38}Ar, ^{37}Ar, and ^{36}Ar) were measured at each temperature step using a Mass Analyzer Products (MAP 215) mass spectrometer at the U.S. Geological Survey, Denver, CO, operated in the static mode, and the isotopic compositions were corrected for mass discrimination. Apparent ages were calculated using the decay constants of Steiger and Jäger (1977). The determination of whether apparent ages in individual age spectra constituted a plateau was made using the criteria of Fleck et al. (1977). Following these criteria, a plateau is defined as comprising two or more contiguous gas fractions that yield apparent ages that are statistically identical at 95% confidence (using the critical value test of Dalrymple and Lanphere (1969)) and which together constitute greater than 50% of the total potassium-derived $^{39}Ar_K$ released in the step-heating experiment. Individual mean ages were calculated using a weighted mean, where weighting is by the inverse of the analytical variance. Calculation of the total gas and plateau dates includes the analytical uncertainty in the determination of the flux parameter *J*.

(Fig. 2A). Sampling was conducted in order to asses the possibility of paleomagnetic differences between the geochemical groups identified by Wooden et al. (1978) and in order to evaluate the validity of the isotopic dates identified by the original Rb–Sr determinations. Most dikes were sampled at more than one site in order to minimize effects of possible lightning strikes, alteration, or variation in magnetic properties on the natural remanent magnetization (NRM) of the samples. Basement rocks adjacent to dikes were sampled at four sites to provide data for baked contact tests (Everitt and Clegg, 1962) to determine if the dike magnetizations are primary or secondary. For comparison purposes, samples from three sites were collected from basement rocks well-removed from the thermal effects of the dikes, and one site was collected from a small Late Cretaceous intrusion.

Between 7 and 16 samples were obtained at each site using a portable drill, with each sample oriented using magnetic and solar compasses and a clinometer. The natural remanent magnetization (NRM) of each specimen was measured using cryogenic or spinner magnetometers. Detailed progressive alternating field (AF) and/or thermal demagnetization was conducted on all samples. Demagnetization results were assessed using orthogonal vector diagrams, stereographic projections,

and normalized magnetic intensity decay plots. Characteristic and secondary components of magnetization, recognized by co-linear demagnetization paths on vector diagrams, were calculated using principal components analysis (Kirschvink, 1980). Where demagnetization failed to isolate a single characteristic direction or stable endpoint but yielded a curvilinear demagnetization trajectory due to overlapping coercivity and/or unblocking temperatures, we calculated remagnetization circles (McFadden and McElhinny, 1988). Site mean directions from sites consisting solely of well-defined linear data were calculated using standard Fisher (1953) statistics; site mean directions based on a combination of linear and great circle data were calculated using the maximum likelihood approach method of McFadden and McElhinny (1988). Magnetic carriers were identified by demagnetization characteristics during AF or thermal demagnetization, acquisition and AF demagnetization of both isothermal remanent magnetization (IRM) and anhysteretic remanent magnetization (ARM) experiments, and by Curie temperature determinations. Curie temperature determinations were made in air in saturating fields of 0.2–0.4 T using an electrobalance similar to that described by Larson et al. (1975). Additionally, low-frequency magnetic susceptibility versus temperature determinations were made in a flowing argon atmosphere using a Kappabridge similar to that described by Hrouda (1995). Petrographic examination of Fe-oxide phases in reflected light using an oil immersion lens was also used to investigate the carriers of magnetization in some dike samples.

4. Paleomagnetic results and interpretation

Ninety-three percent of the Tobacco Root dike samples gave interpretable results after AF or thermal demagnetization, and three distinct magnetizations were identified. Two of these magnetizations correspond to dikes represented by geochemical groups A and B, identified by Wooden et al. (1978); however, to distinguish the magnetization groups from the geochemical groups, we term these the A_1 and B_1 magnetizations. As discussed below, we consider these to be primary thermoremanent magnetizations (TRMs) that record two distinct episodes of dike emplacement during the Proterozoic. An additional magnetization, termed the *R* magnetization, is probably a Late Cretaceous or early Tertiary secondary remanence that completely overprinted primary dike magnetizations at some localities. The small percentage of samples that did not yield interpretable results are characterized by erratic directional behavior or intensity changes during demagnetization, or were dominated by low coercivity magnetizations of inconsistent orientation. Because many of these samples had NRM intensities greater than 15 A/m, we suspect that they have been struck by lightning. Data from these samples will not be considered further.

4.1. A_1 Magnetization

Samples from 24 sites in the southern Tobacco Root Mountains and one site in the northern Tobacco Root Mountains give characteristic remanent magnetizations (ChRM) of south and southwest declination (*D*) and moderate to steep positive inclination (*I*) or north and northeast declination and moderate to steep negative inclination. For the purposes of this discussion, we refer to these as normal and reverse polarity magnetizations but note that these terms refer to present-day coordinates only and make no inference regarding the hemisphere in which the magnetization was acquired. This magnetization is found in dikes that belong to geochemical groups A and C of Wooden et al. (1978), and in some previously unmapped dikes in both the northern and southern Tobacco Root Mountains.

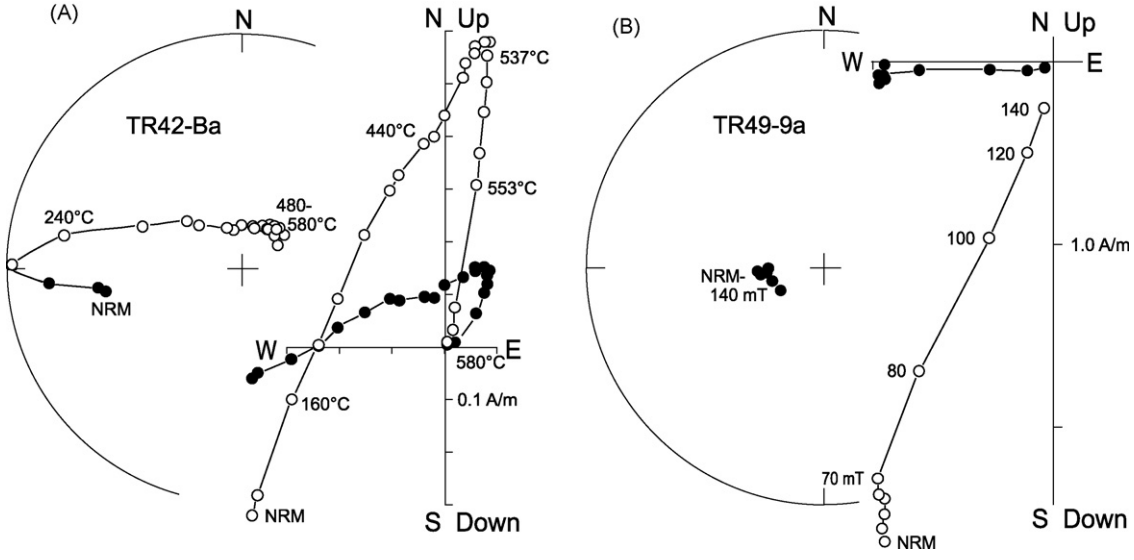


Fig. 4. In situ orthogonal vector diagrams and equal-area projections showing the behavior of paleomagnetic samples from group A dikes during thermal and AF demagnetization. Demagnetization results from individual sites from group A dikes show discrete antipodal magnetizations that are well defined during the demagnetization experiments; this magnetization is termed the A_1 remanence. For the equal area projections, solid symbols are projections on the lower hemisphere and open circles are projections on the upper hemisphere. For the vector diagrams, solid circles are projections onto the horizontal plane, and open circles represent projections on the vertical plane.

Table 2
Paleomagnetic site- and dike-mean data from Proterozoic mafic dikes in the Tobacco Root Mountains, southwestern Montana

No.	Site	Slat.	Slong.	$N_T/N_S/N_R$	D	I	k	α_{95}	Plat.	Plong.
A_1 magnetization										
1	TR5	45.4187°N	112.1518°W	9/9	242.4°	68.6°	101.0	5.2°	21.1°N	212.0°E
2	TR4	45.4192°N	112.1511°W	9/8	30.6°	−55.7°	20.8	12.4°	3.8°S	223.6°E
	TR51	45.4197°N	112.1558°W	9/9	13.0°	−69.8°	93.1	5.4°	9.7°N	240.1°E
	TR52	45.4195°N	112.1549°W	7/7	358.0°	−59.2°	201.4	4.3°	4.6°S	249.4°E
Dike mean				3 sites	14.6°	−62.3°	54.4	16.9°	0.1°	237.4°E
3	TR41	45.4198°N	112.1492°W	8/7	38.9°	−61.4°	125.2	7.2°	4.5°N	220.2°E
	TR42	45.4209°N	112.1561°W	8/8	37.6°	−66.8°	244.3	3.6°	10.3°N	224.0°E
Dike mean				15 samples	38.3°	−64.3°	150.2	3.1°	7.5°N	222.2°E
4	TR12	45.4220°N	112.1142°W	10/9	271.4°	70.3°	205.5	3.6°	36.1°N	201.8°E
	TR13	45.4223°N	112.1141°W	8/0	Unstable—no stable components isolated					
5	TR32	45.4341°N	112.1199°W	7/4/2	36.7°	−57.9°	53.2	9.7°	0.2°N	220.0°E
	TR34	45.4336°N	112.1176°W	9/9	19.4°	−63.8°	230.5	3.4°	2.5°N	234.4°E
Dike mean				15 samples	27.0°	−61.7°	75.5	4.4°	1.5°N	228.4°E
6	TR49	45.4165°N	112.1243°W	10/10	254.8°	67.7°	181.5	3.6°	25.7°N	205.1°E
	TR35	45.4151°N	112.1216°W	13/12	258.6°	61.8°	27.3	8.5°	22.6°N	197.0°E
Dike mean				22 samples	257.1°	64.5°	43.5	4.8°	24.0°E	200.4°E
7	TR57	45.4169°N	112.1271°W	8/8	251.9°	55.1°	66.3	6.9°	13.7°N	195.2°E
	TR58	45.4174°N	112.1291°W	9/8	256.9°	51.9°	42.0	8.6°	14.4°N	189.9°E
Dike mean				16 samples	254.5°	53.5°	52.9	5.1°	14.1°N	192.5°E
8	TR36	45.4273°N	112.0842°W	9/3/4	60.8°	−74.1°	16.7	16.2°	26.7°N	219.0°E
9	TR43	45.362°N	111.802°W	9/0	Unstable—no stable components isolated					
	TR44	45.362°N	111.797°W	10/6	36.7°	−48.2°	102.8	6.6°	8.3°S	216.4°E
10	TR54	45.360°N	112.935°W	11/7/2	44.2°	−60.8°	70.1	6.3°	5.7°N	215.6°E
11	TR18	45.3873°N	112.0519°W	12/5/5	37.9°	−58.6°	38.9	8.1°	1.3°N	219.6°E
	TR47	45.3894°N	112.0550°W	13/12	21.5°	−48.5°	55.4	5.9°	12.6°S	228.9°E
Dike mean				22 samples	28.0°	−53.3°	35.7	5.3°	6.8°S	224.8°E
12	TR60	45.409°N	112.061°W	15/14	224.9	56.9	82.6	4.4	2.2°N	213.8°E
13	TR46	N.A.	N.A.	18/11/4	13.7°	−35.2°	27.1	7.5°	23.7°S	233.8°E
14	TR17	45.3950°N	112.0592°W	11/0	Contains a K/T overprint that obscures the primary remanence					
	TR73	45.3955°N	112.0376°W	11/10	249.9	62.8	118.0	4.5	18.9°N	202.6°E
B_1 magnetization										
15	TR14	45.4238°N	112.1222°W	9/8	311.4°	−21.4°	14.5	15.1°	18.6°N	118.8°E
16	TR6	45.4259°N	112.1344°W	12/3/8	297.5°	−11.8°	45.0	7.2	14.4°N	133.5°E
	TR7	45.4261°N	112.1340°W	15/3/10	294.7°	−9.1	47.9	6.3°	13.6°N	136.6°E
	TR15	45.4231°N	112.1138°W	8/0	Unstable—no stable components isolated					
	TR16/50	45.4229°N	112.1127°W	15/15	291.7°	−22.1°	72.3	4.5°	6.5°N	134.3°E
	TR30	45.4263°N	112.1335°W	10/4/3	297.8°	−7.5°	34.2	10.9°	16.3°N	134.7°E
	TR31	45.4253°N	112.1319°W	9/3/6	300.8°	−8.9°	54.5	7.5°	17.6°N	131.9°E
Dike mean					296.6°	−11.9°	142	6.4°	13.8°N	134.2°E
17	TR8	45.4160°N	112.1194°W	9/2/5	300.6°	−6.1°	206	4.7	20.9°N	135.1°E
	TR9	45.4161°N	112.1191°W	7/1/5	307.6°	−15.8°	67.7	9.7	18.9°N	123.9°E
Dike mean					304.0°	−10.9°	67.6	4.9°	18.8°N	128.6°E
18	TR10	45.4204°N	112.1137°W	11/3/7	297.8°	−9.3°	122	4.6°	15.6°N	134.1°E
	TR11	45.4209°N	112.1136°W	9/9	293.5°	−21.9°	31.2	9.4°	7.7°N	133.0°E
	TR38	45.4206°N	112.1121°W	9/4/5	297.9°	−21.7°	138	4.6	10.6°N	129.7°E
Dike mean				3 sites	296.4°	−17.7°	114	11.6°	11.3°N	132.3°E
19	TR37	45.43345°N	112.1124°W	8/4/4	292.3°	−9.3°	63.6	7.3°	10.6°N	129.7°E
20	TR40	45.4036°N	112.0524°W	12/4/8	316.4°	−21.1°	36.1	7.6	21.4°N	114.6°E
	TR48	45.380°N	111.987°W	13/4/8	298.3°	−9.3°	75.0	5.3°	15.9°N	133.9°E

Table 2 (*Continued*)

No.	Site	Slat.	Slong.	$N_T/N_S/N_R$	D	I	k	α_{95}	Plat.	Plong.
Dike mean					307.0°	−15.3°	27.1	5.8°	18.8°N	124.7°E
21	TR19	45.369°N	111.928°W	10/4/6	304.3°	−14.7°	46.6	7.3°	17.5°N	127.2°E
	TR20	45.368°N	111.927°W	7/0	Contains a K/T overprint that obscures the primary remanence					
	TR21	45.368°N	111.927°W	10/3/5	303.2°	−17.9°	98.7	6.0°	15.4°N	127.1°E
Dike mean				18 samples	303.8°	−17.9°	62.6	4.4°	15.8°N	126.6°E
22	TR45	45.350°N	111.828°W	16/3/10	295.1°	−23.0°	23.7	9.0°	8.3°N	131.7°E
23	TR22	45.396°N	111.878°W	11/6	307.8°	−12.2°	21.3	14.8°	20.6°N	125.2°E
	TR61	45.3957°N	111.8748°W	16/4/9	308.4°	−21.9°	16.3	10.9°	16.7°N	121.4°E
Dike mean				20 samples	308.2°	−18.8°	17.5	8.3	18.0°N	122.7°E
24	TR62	45.3689°N	111.9274°W	17/7/11	294.8°	−20.2°	25.7	7.1°	9.3°N	132.9°E
25	TR71	45.4229°N	112.1062°W	9/7	303.9°	−22.9°	147.6	5.0	13.7°N	124.6°E
R magnetization										
26	TR6	45.4259°N	112.1344°W	13/5	340.6°	56.3°	152.2	6.2°	73.1°N	134.2°E
	TR30	45.4263°N	112.1335°W	8/2	0.0°	65.4°	—	—	87.9°N	247.9°E
Dike mean				7 samples	345.0°	59.2°	91.6	6.3°	77.7°N	136.9°E
27	TR8	45.4160°N	112.1194°W	11/3	330.8°	54.6°	104.2	12.1°	65.6°N	143.2°E
	TR9	45.4161°N	112.1191°W	7/1	341.2°	46.2°	—	—	66.7°N	114.2°E
Dike mean				4 samples	333.8°	52.6°	92.9	9.6°	66.5°N	135.7°E
28	TR17	45.3950°N	112.0592°W	11/9	317.2°	64.7°		8.8°	60.6°N	175.9°E
29	TR20	45.368°N	111.927°W	7/4	350.7°	74.0°	150.7	7.5°	74.2°N	229.9°E
	TR21	45.368°N	111.927°W	11/1	337.6°	50.9°	—	—	67.8°N	126.4°E
Dike mean				5 samples	348.6°	69.3°	49.6	11.0°	79.4°N	206.6°E
30	TR45	45.350°N	111.828°W	16/2	329.3°	56.4°	—	—	65.6°N	148.0°E
31	TR55	45.680°N	112.039°W	13/11	335.3°	73.1°	30.3	8.4°	70.2°N	208.1°E
	TR56	45.681°N	112.041°W	12/11	7.6°	70.0°	90.8	4.8°	80.4°N	275.8°E
Dike mean				22 samples	2.0°	71.6°	45.6	4.6°	79.2°N	253.9°E
32	MR2	45.5669°N	111.6975°W	11/11	9.9°	71.9°	123.8	7.3°	77.2°N	273.4°E
Cretaceous/Tertiary intrusion										
33	TR39	45.4215°N	112.0534°W	8/7	191.0°	−68.1°	110.1	5.8°	80.7°N	295.6°E

Note: N_T is the number of samples measured; N_S is the number of samples yielding a stable end point used in the estimate of the site-mean direction; N_R is the number of samples that gave remagnetization circles used in the estimate of the site-mean direction (if used); D and I are declination and inclination of the site-mean direction; k is the best estimate of the precision parameter (Fisher, 1953); α_{95} is the semi-angle of the cone of 95% confidence about the site mean direction; Plat. and Plong. are the latitude and longitude of the virtual geomagnetic pole. The VGPs for dikes with the A_1 remanence were calculated after inversion to a southwest declination and positive inclination. K/T is Cretaceous/Tertiary. MR2 is a paleomagnetic site in a northwest-striking Proterozoic mafic dike from the Madison Range that is suspected of having been remagnetized during Late Cretaceous/Tertiary time.

NRM directions from these samples are scattered, but largely define a plane that contains both normal and reverse polarity magnetizations and the direction of the present day field (i.e. $D = 13°$, $I = 70°$). NRM intensities range from 18.6×10^{-3} to 19.9×10^{1} A/m, with a geometric mean of 1.76 A/m. Most samples contained two magnetic components. The first magnetization removed is usually of north declination and moderate positive inclination; a mode calculated from the overprint directions is $D = 1.3°$, $I = 75.8°$ ($\beta_{95} = 9.4°$), which is essentially identical to the direction of the present-day field. This remanence is usually removed by AF demagnetization of 5–20 mT or by temperatures less than 350 °C. The direction and demagnetization characteristics indicate that this is probably a viscous remanent magnetization (VRM) acquired in the present-day field. Other low coercivity/low laboratory unblocking temperature overprints are inconsistent in orientation and their origin is unclear; an exception is discussed below.

Following removal of secondary magnetizations, 72% of the samples from this group showed behavior in which the paleomagnetic vector decayed linearly to the origin of their vector diagram during AF or thermal demagnetization (Fig. 4B). Such behavior usually indicates the isolation of a single demagnetization component, which we consider to be the sample ChRM ("stable endpoint"). Directions for the stable endpoints were well defined by principal components analysis (PCA), as indicated by maximum angular deviation (MAD) values of 2° to 11°, with 91% ≤7°. The stable endpoints define two approximately antipodal clusters. Four percent of the samples failed to yield stable endpoints, but gave curvilinear demagnetization paths that converged toward either the normal or reverse polarity A_1 direction during demagnetization. This behavior is caused by the presence of two magnetizations of similar coercivity and/or laboratory unblocking temperature spectra such that a stable endpoint could not be isolated. For these samples, magnetiza-

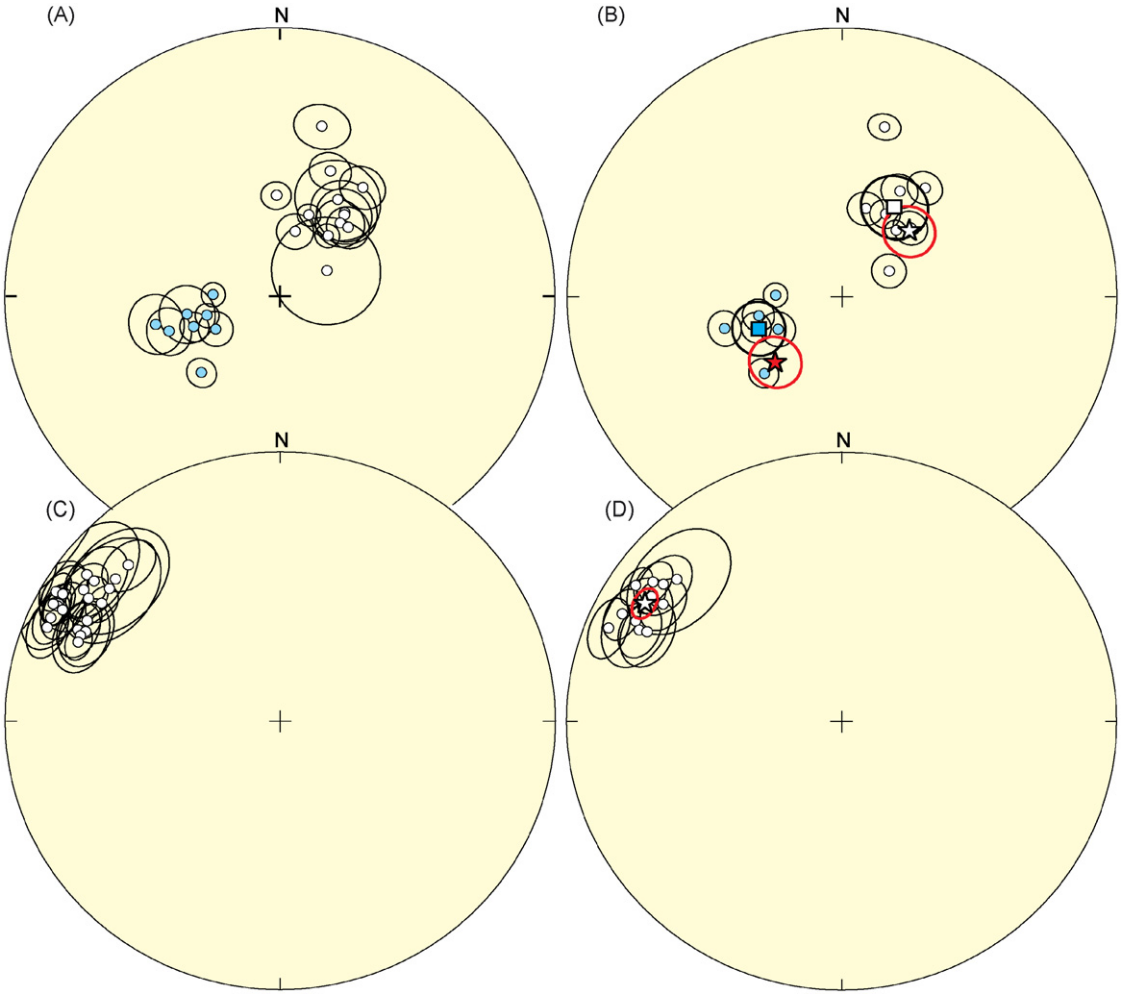


Fig. 5. Equal area projections showing in situ paleomagnetic results from Proterozoic mafic dikes from the southern Tobacco Root Mountains, Montana. (A) Individual paleomagnetic site-mean directions and α_{95} cones of confidence from group A dikes. (B) Dike mean directions and α_{95} cones of confidence from the group A dikes that show the normal and reverse polarity A_1 magnetizations. Also shown are the means of the normal and reverse polarity group mean directions (filled and open squares, respectively) and cones of confidence. A grand-mean direction based on inversion of all sites to a common polarity is shown by the stars for normal and reverse polarity coordinates. (C) Individual paleomagnetic site-mean directions and α_{95} cones of confidence from group B dikes showing the B_1 magnetization. (D) Dike mean directions and α_{95} cones of confidence from the group B dikes and the overall grand-mean direction (star).

Table 3
Summary of paleomagnetic data from Proterozoic mafic dikes in the Tobacco Root Mountains, southwestern Montana

	D	*I*	*k*	α_{95}	*N*	Plat.	Plong.	*K*	A_{95}
Group A normal polarity-in situ	248.7°	63.5°	71.6	8.0°	6	19.6°N	204.0°E	36.2	11.3°
Group A reverse polarity-in situ	30.0°	−58.2°	33.8	9.7°	8	0.3°N	222.8°E	23.0	12.8°
Group A mean combined-in situ	225.0°	61.8°	27.9	7.7°	14	8.7°N	216.1°E	15.8	10.3°
Group B mean-in situ	301.5°	−17.1°	117	4.0°	12	14.6°N	127.0°E	189.5	3.2°
R magnetization-TR dikes only	337.5°	62.9°	65.7	8.3°	6	74.6°N	164.2°E	33.4	11.8°
R magnetization-TR dikes + KTi	346.8°	64.2°	46.6	8.2°	8	81.2°N	169.3°E	21.8	12.1°

Note: *D* and *I* are declination and inclination of the group-mean directions; *k* is the best estimate of the precision parameter (Fisher, 1953); α_{95} is the semi-angle of the cone of 95% confidence about the site mean direction; Plat. and Plong. are the latitude and longitude of the paleomagnetic pole; *K* is the best estimate of the Fisher mean of the paleomagnetic pole; A_{95} is the semi-angle of 95% confidence of the paleomagnetic pole; KTi = Cretaceous/Tertiary intrusion.

tion planes were calculated using PCA from four or more points along the demagnetization path. All planes used in the analysis had MAD values $\leq 13^\circ$, with 80% $\leq 5^\circ$.

Site-mean directions are generally well grouped as evidenced by Fisher (1953) k values that ranged from 17 to 244 and α_{95} values of 3.6–16.2° (Table 2; Fig. 5A). Where multiple sites were collected from the same dike, the dike mean direction was calculated by averaging site-mean directions (three or more sites) or by averaging sample directions (only two sites). This gave what we interpret to be 14 independent recordings of the geomagnetic field (Fig. 5B). The reverse and normal group-mean directions ($D = 248.7^\circ$, $I = 63.5^\circ$, $k = 71.6$, $\alpha_{95} = 8.0^\circ$, $N = 6$ dike

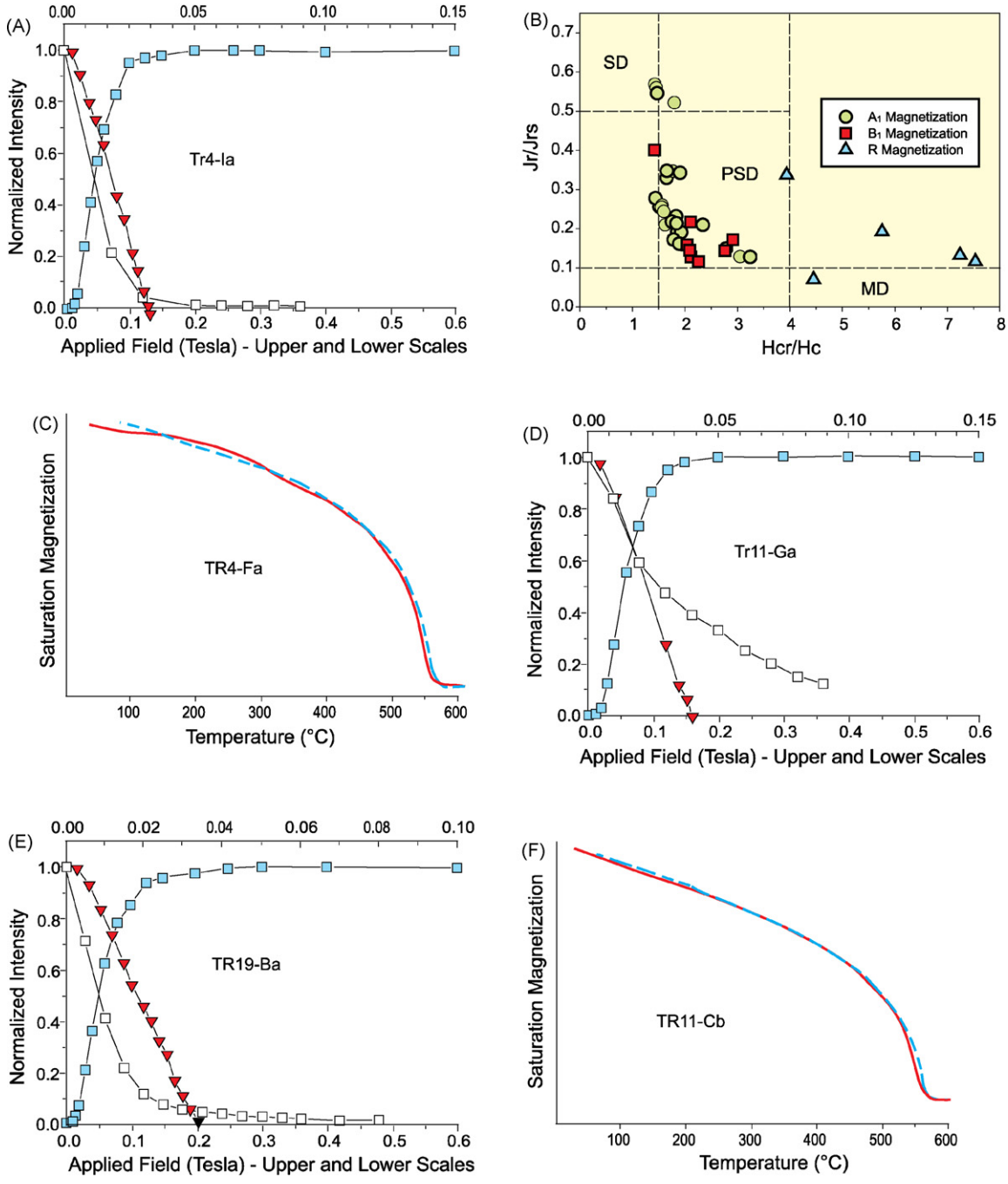


Fig. 6. Rock magnetic data from dikes from the southern Tobacco Root Mountains. (A) Normalized curve showing the acquisition of an isothermal remanent magnetization (IRM) (shaded squares) in a dike sample that shows the A_1 magnetization in an applied dc field (lower scale), AF demagnetization of the natural remanent magnetization (open square) (upper scale), and backfield dc demagnetization of the IRM (filled triangle) (upper scale). (B) Plot showing the ratio of saturation remanence (Jr) to saturation magnetization (Js) vs. the ratio of coercivity of remanence (Hcr) to coercivity (Hc) (Day et al., 1977) for samples that show the A_1, B_1 and R magnetizations. (C) Strong field thermomagnetic curve for a sample that shows the A_1 magnetization. Solid line represents the heating curve; the broken line illustrates the cooling curve. (E) IRM acquisition and demagnetization curves for a group B dike sample in which the B_1 magnetization is well defined. (F) IRM acquisition and demagnetization curves for a group B sample in which the B_1 sample is obscured by a Late Cretaceous/early Tertiary overprint. (F) Strong field thermomagnetic curve from for a sample that shows the B_1 magnetization.

means, and $D = 30.0°$, $I = -58.2°$, $k = 33.8$, $\alpha_{95} = 9.7°$, $N = 8$ dike means) (Table 3), after inversion to a common normal polarity, have an angular difference of $18.4 \pm 8.8°$ using the method of Debiche and Watson (1995). The similarity in directions indicates that remanence acquisition must have spanned at least two polarity chrons, but overall the data do not yield a positive reversal test using the criteria of McFadden and McElhinny (1990) (i.e. $\gamma_o = 19.3°$; $\gamma_c = 12.4°$). Combining the normal and reverse polarity site-mean directions yields a grand-mean direction of $D = 45.0°$, $I = -61.8°$ ($k = 27.9$, $\alpha_{95} = 7.5°$, $N = 14$) (Fig. 5B; Table 2).

Paleomagnetic and rock magnetic evidence indicates that the A_1 remanence is carried by magnetite. In general, the reverse polarity A_1 samples are characterized by low to moderate (10–30 mT) median destructive fields (MDFs) and distributed to discrete laboratory unblocking temperatures. Complete unblocking occurred at temperatures below 580 °C, the Curie temperature of pure magnetite. Normalized intensity curves show an initial exponential decay of the magnetic moment during AF demagnetization (Fig. 6A), followed by gradual decay at higher fields. Saturation of IRM occurs in fields less than 200 mT indicating that the remanence is dominated by a low coercivity phase. The ChRM of these samples is probably carried by a combination of multidomain and single domain magnetite (Fig. 6B). In contrast, normal polarity samples usually have greater MDFs (i.e. 30–60 mT), have less well defined secondary magnetizations, and thermally discrete or "knee-shaped" laboratory unblocking temperatures less than 580 °C. These samples are commonly more altered than samples that contain the reverse polarity ChRM. Their normalized AF decay curves are of single-domain form (Dunlop and Özdemir, 1997). Strong-field thermomagnetic determinations from normal and reverse polarity samples are essentially identical and give reversible heating and cooling curves (Fig. 6C) with a single Curie temperature at 568 ± 8°C (1S.D.; $n = 5$ samples). The paleomagnetic and rock magnetic characteristics of these samples indicate that low-Ti titanomagnetite carries the A_1 magnetization in these rocks.

Two baked contact tests were conducted on dikes with the A_1 magnetization in order to determine whether the A_1 magnetization is a primary thermoremanent magnetization (TRM) or a secondary magnetization acquired at some later time. The first test is from samples collected within 0.3 m of an 8-m wide dike that has a ChRM of $D = 14.6°$ and $I = -62.3°$ ($k = 54.4$, $\alpha_{95} = 16.9°$, $N = 3$ sites; dike 2, Table 2). The quartzofeldspathic gneiss samples have well-defined secondary magnetizations of low coercivity and laboratory unblocking temperatures that are readily removed during demagnetization (Fig. 7B). The secondary magnetization is of north declination and moderate positive inclination with a site-mean direction of $D = 2.5°$ and $I = 76.6°$ ($k = 504.6$, $\alpha_{95} = 2.5°$ $n = 8$ samples). Because of its ready removal during demagnetization and its normal polarity direction, this magnetization is probably a VRM acquired in the present-day or Recent geomagnetic fields. With further demagnetization, these samples give a magnetization of north to northeast declination and moderate to steep negative inclination. A site-mean direction from this component gives $D = 34.8°$ and $I = -61.7°$ ($k = 51.7$, $\alpha_{95} = 7.2°$, $n = 8$ samples), which is essentially identical to the remanence of the dike (Fig. 7A and Fig. 8A). The similarity of directions between the dike and host gneiss samples strongly suggests that the dike has baked the adjacent gneiss to form a TRM and that the dike magnetization is primary and was acquired during cooling following dike emplacement. AF and thermal demagnetization characteristics, particularly the complete unblocking of the magnetization at temperatures below 580 °C, indicate that the magnetization in the baked contact samples is carried by magnetite.

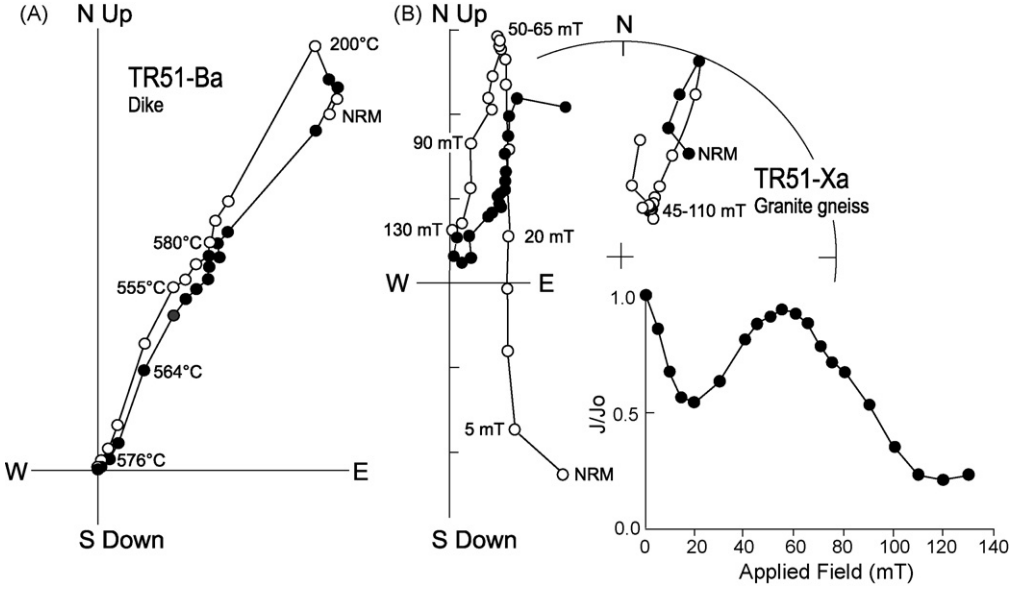


Fig. 7. In situ paleomagnetic baked contact results from a group A dike. (A) Orthogonal vector diagram from dike sample showing a well defined characteristic direction of northeast declination and moderate negative inclination. (B) Orthogonal vector diagram, equal area projection, and magnetic intensity vs. applied field showing demagnetization characteristics of a sample of granite gneiss adjacent in the contact zone adjacent to the dike. Gneiss samples from this contact test show well-resolved characteristic directions identical to that of the dike.

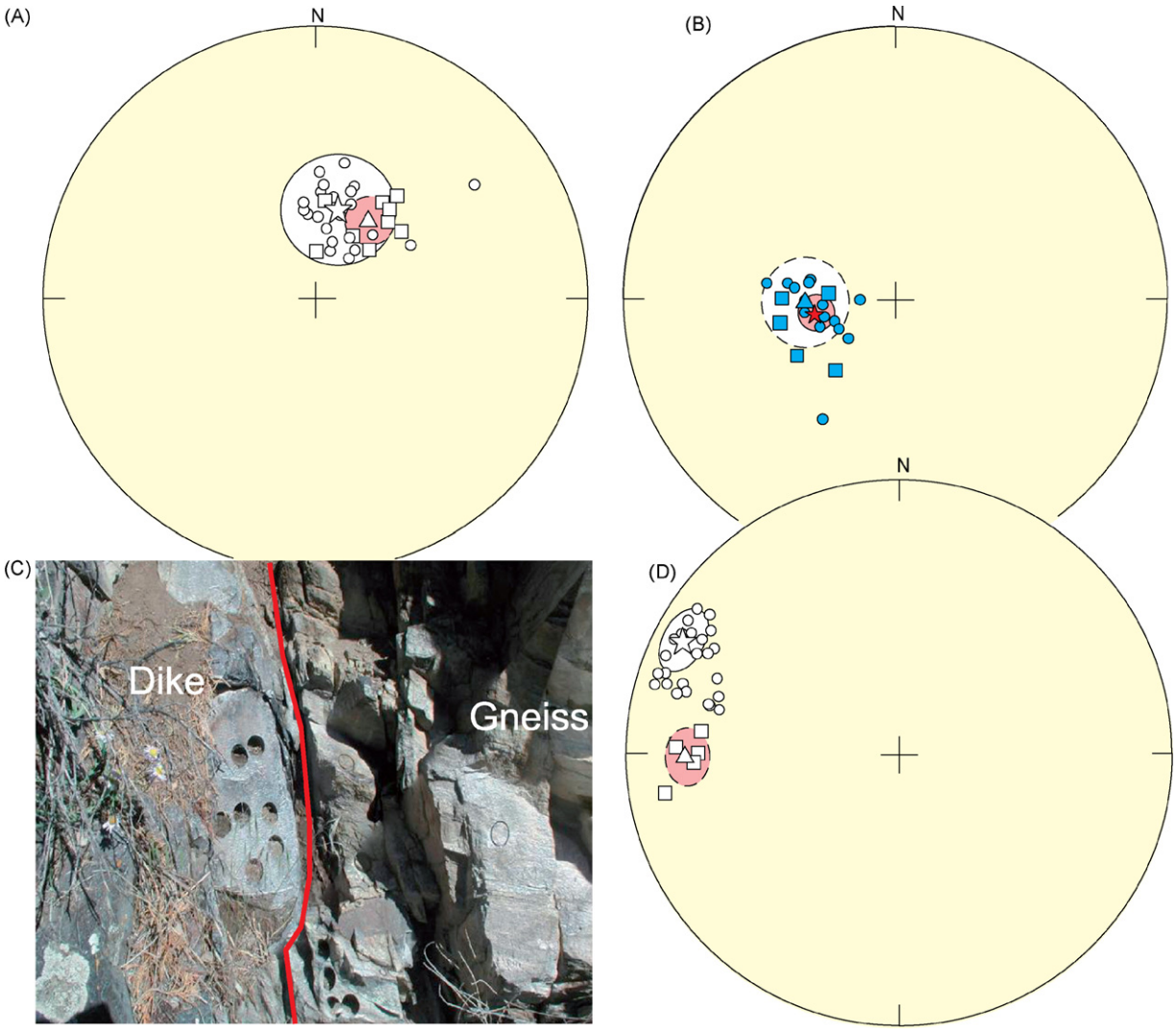


Fig. 8. Baked contact tests from three dikes from the southern Tobacco Root Mountains. (A) Baked contact test from a group A dike with a mean direction of east and northeast declination and moderate to steep negative inclination. The baked contact tests from the group A dikes are positive and indicate that both the normal and reverse polarity directions are primary thermoremanent magnetizations acquired during subsolidus cooling following dike emplacement. (B) Baked contact test from a group A dike with a mean direction of west and southwest declination and moderate to steep inclination. Circles show characteristic directions from two dike sites; star shows the mean direction from the dike; squares are the characteristic directions obtained from gneiss samples adjacent to the dike contact; triangle shows the mean direction obtained from the dike samples; α_{95} cones of confidence are also shown for the mean directions. (C) Photo showing chilled dike margin and contact zone of a group B dike. Samples for a baked contact test were collected from the dike margin for both paleomagnetic and $^{40}Ar/^{39}Ar$ dating purposes. Paleomagnetic drill holes shown in the photograph have a diameter of 2.9 cm. (D) Baked contact test from a group B dike shown in photo. Circles are characteristic directions obtained from the dike and squares are from gneiss samples within 10 cm of the dike contact. The dike and contact gneiss yield similar characteristic directions of northwest and west declination and shallow negative inclination. Note that the mean directions of the dike and contact samples are distinct at the 95% confidence level, suggesting that the baked contact test is negative. However, $^{40}Ar/^{39}Ar$ dating of biotite from one of the host samples yields an apparent age of 787 Ma, which is consistent with the apparent age of the dikes as determined by U–Pb and $^{40}Ar/^{39}Ar$ dates on hornblende from the group B dikes. We suggest that the apparent discordance of the dike and gneiss directions may be due to deflection of the magnetic vector by the strong metamorphic fabric of the gneiss samples. The similarity of directions and the concordance of the biotite $^{40}Ar/^{39}Ar$ date from these samples suggests that the baked contact test should be considered to be positive and that the B_1 magnetization is a thermoremanent magnetization acquired during cooling following emplacement of the group B dikes.

The second baked contact test is from a 5-m wide dike, which has a characteristic direction of southwest declination and steep positive inclination. At this locality, we collected 12 samples along a traverse perpendicular to the dike contact out for 8 m. Gneiss samples from within a meter of the dike yield magnetizations that are essentially identical to that of the dike (Fig. 9A–C). Five samples collected within a meter of the dike give a mean direction of $D = 248.9°$, $I = 60.0°$ ($k = 40.4$, $\alpha_{95} = 12.2°$), which is essentially identical to the dike mean direction of $D = 257.1°$, $I = 64.5°$ ($k = 43.5$, $\alpha_{95} = 4.8°$, $n = 22$ samples; dike 6, Table 2) (Fig. 8B). At greater distances from the contact, magnetizations in the gneiss are either randomly oriented, directionally unstable, or of low coercivity and similar to the present-day field direction (Fig. 9D). The baked contact results from this site strongly suggest gneiss magnetization was caused by thermal affects accompanying dike emplacement. Rock magnetic characteristics indicate that magnetizations in baked and unbaked gneiss samples are carried by magnetite.

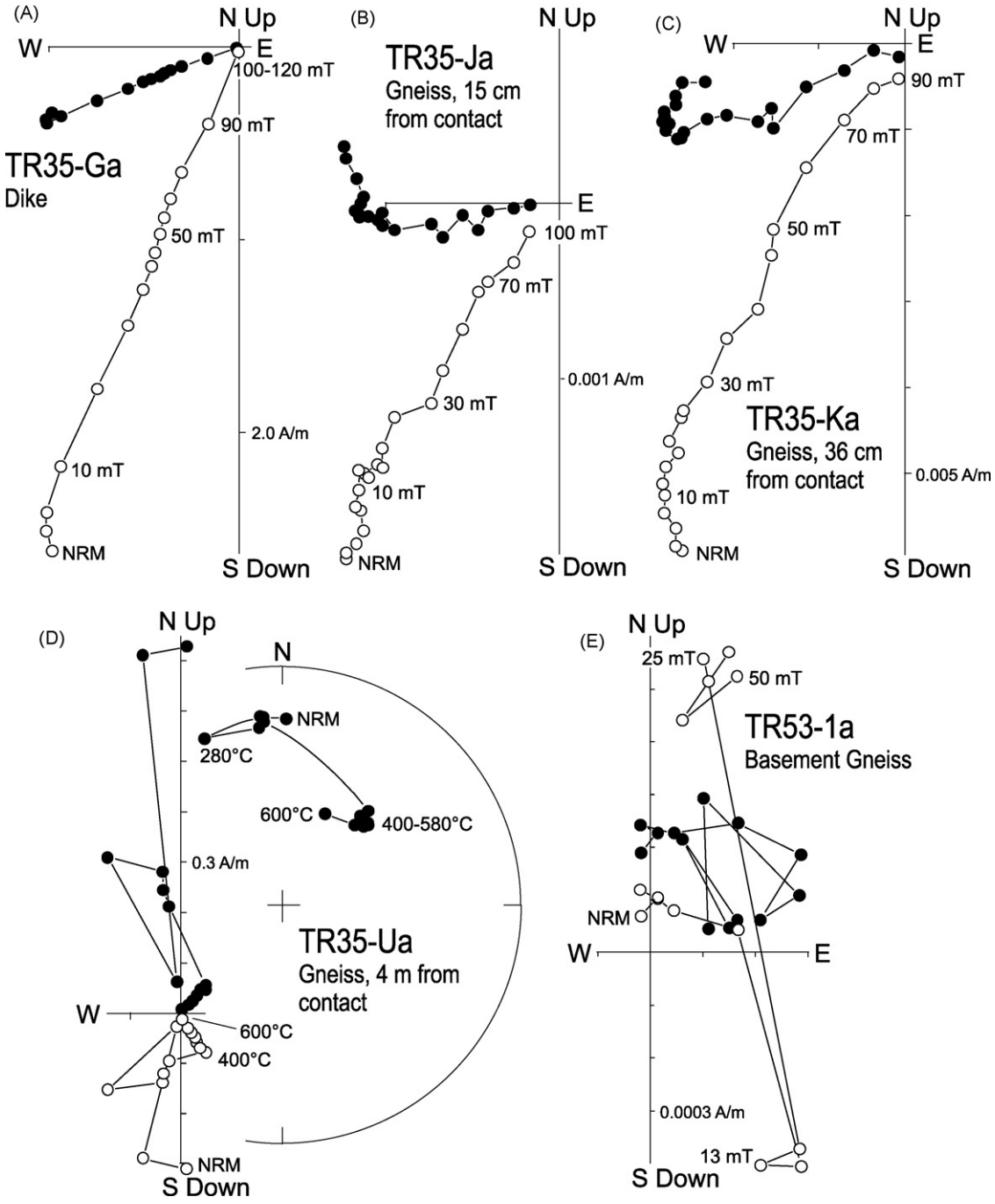


Fig. 9. Paleomagnetic baked contact results from a group A dike. (A) Orthogonal vector diagram from dike sample showing a well defined A_1 characteristic direction of southwest declination and moderate to steep positive inclination. (B and C) Samples of basement gneiss adjacent to the dike contact at distances of 16 and 36 cm, respectively. Both gneiss samples show well-resolved characteristic A_1 directions identical to that of the dike. (D) Orthogonal vector diagram and equal area projection showing thermal demagnetization behavior of a gneiss sample collected 4 m from the dike contact. The characteristic direction from this sample is of north declination and moderate positive inclination and is distinct from the magnetizations isolated in the dike and samples adjacent to the dike contact. (E) Typical demagnetization of basement gneiss samples that are well removed from the thermal effects of nearby Proterozoic or Tertiary intrusions. All diagrams are in situ coordinates.

The presence of antipodal baked contact tests strongly suggests that the magnetization in both contact zones was caused by thermal affects associated with dike emplacement and that the observed dike normal and reverse polarity magnetizations are primary TRMs. Unfortunately, although these tests suggest a primary origin for the A_1 remanence, they are not fully conclusive because we cannot demonstrate that the dike magnetizations are superimposed on geologically stable, preexisting magnetizations in the gneiss. However, we note that the gneiss samples well removed from the thermal effects of the Proterozoic dikes typically do not yield coherent magnetizations of any geologic age (Fig. 9E).

4.2. B_1 Magnetization

Ninety-one percent of samples from group B dikes gave interpretable and usable paleomagnetic results. NRM directions are dominated by magnetizations of north declination and moderate positive inclination, although there is significant streaking towards northwest declinations and shallow inclinations. The range of NRM intensities (8.2×10^{-5} to 14.8×10^{1} A/m) is similar to that of the dikes with the A_1 magnetization, but the mean is significantly lower (i.e. 7.97×10^{-1} A/m). Following demagnetization, paleomagnetic samples from the group B dikes gave results that can be divided into two types: (1) samples in which a well-defined, univectorial magnetization of northwest declination and shallow negative inclination, the B_1 remanence, is readily isolated (40% of samples); (2) those in which the B_1 magnetization is partially to almost completely dominated by secondary magnetizations such that the characteristic direction could not be fully isolated (43.6% of samples); and (3) those that are completely dominated by a magnetization of north declination and moderate inclination, termed the *R* magnetization, discussed in a subsequent section (7.5% of samples).

Samples in which the B_1 remanence is well defined are typically from fine-grained (<1 mm diameter) phases or chilled zones of the group B dikes. Secondary overprints, where present, are minor and easily removed by AF or thermal demagnetization (Fig. 10A and B). The B_1 remanence in these samples is characterized by moderate to high MDFs (30–70 mT). Normalized intensity decay curves from thermally demagnetized samples are usually "knee-shaped" indicating a discrete range of laboratory unblocking temperatures with 60–80% of the NRM unblocked between 520 and 580 °C. IRM acquisition experiments and AF demagnetization of NRM and IRM behavior and hysteresis data (Fig. 6B and D) suggest that this magnetization is carried by a combination of single- and/or pseudosingle domain magnetite. Strong-field thermomagnetic curves are reversible and give a single Curie point at 565 ± 10 °C ($n = 3$ samples), indicative of low-Ti titanomagnetite (Fig. 6F). Stable endpoints obtained from these samples are well grouped about a northwest, moderate negative inclination direction. We consider this direction to be the ChRM of the group B dikes.

Isolation of the B_1 magnetization in other group B dike samples was somewhat problematic. Most samples are dominated

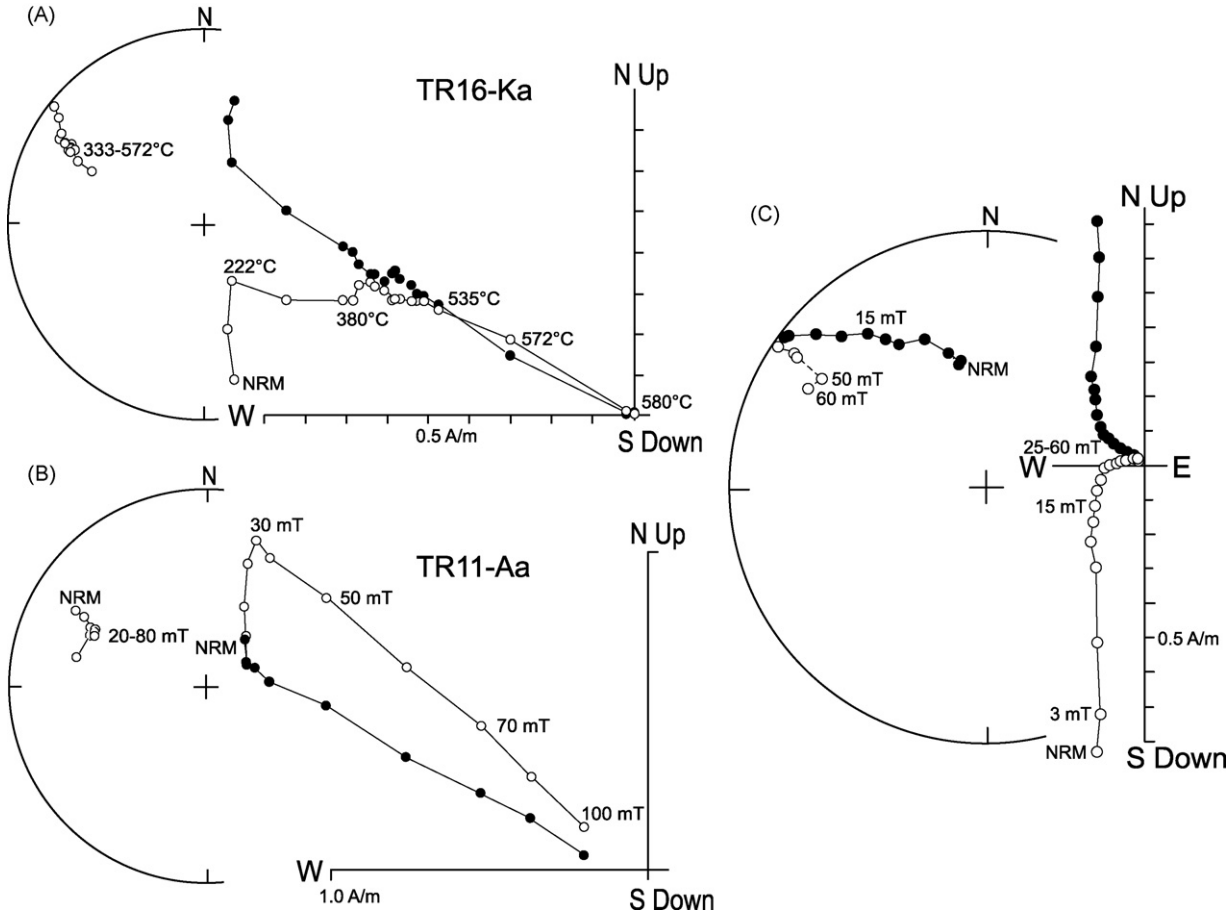


Fig. 10. (A and B) Orthogonal vector diagrams and equal-area projections showing the behavior of paleomagnetic samples from group B dikes that yield well-defined univectorial decay of the B_1 magnetization vector during thermal and AF demagnetization. (C) Vector diagram and equal-area projection of a sample from a group B dike in which the characteristic remanent magnetization could not be isolated due to the overlap of coercivity and unblocking temperature spectra with a magnetization of north declination and moderate positive inclination. For most samples which displayed such behavior, the characteristic remanent magnetization could not be isolated in AF or thermal demagnetization as the remanence became erratic at high AF fields or temperatures. For such samples, remagnetization circle analysis was applied as described in the text. Conventions as in Fig. 4.

by well-defined secondary magnetizations of north declination and moderate positive inclination (Fig. 10C). This overprint is characterized by low to moderate coercivities and distributed laboratory unblocking temperatures. Up to 60% of the NRM is usually removed by 15–20 mT, or by 250–300 °C. A mode based on 157 overprint directions is $D = 356.6°$, $I = 64.1°$ ($\beta_{95} = 3.1°$), which is similar to both latest Cretaceous/early Tertiary magnetizations and the time-averaged and present-day field directions for the site area. It is possible that the observed magnetization is a VRM of recent origin, but partial remagnetization by Cretceous or Tertiary thermal events cannot be ruled out. At higher demagnetization levels, the secondary magnetization is gradually removed, with the resultant remanence directions defining a great circle path toward the B_1 direction defined by the stable endpoints described above. Due to the overlap of the coercivity and unblocking temperature spectra of these magnetizations, and the common onset of erratic directional and intensity changes at peak fields greater than 60 mT, the ChRM of many of these samples is rarely fully isolated. The erratic behavior is probably caused by the acquisition of a spurious, laboratory induced anhysteretic remanent magnetization during AF demagnetization. Most samples that show this behavior are from coarse-grained, dike interiors and contain large (up to 0.5–1.0 mm) titanomagnetite-ilmenite grains. Thermal and low temperature demagnetization methods were similarly ineffective in isolating the ChRM of many samples. Most thermally demagnetized samples became unstable at temperatures above 350 °C. Sulfide minerals are common in these rocks, and this behavior may reflect their thermal breakdown to create new magnetic phases during the demagnetization experiments. Most data used in the calculation of site-mean directions from these rocks are based on AF results.

Paleomagnetic and rock magnetic characteristics of the coarse-grained group B dikes indicate that their magnetization is carried by a combination of multidomain and single-domain or pseudosingle-domain magnetite. AF demagnetization curves show rapid decay of NRM at low fields, followed by gradual decay of the NRM at higher fields. Saturation of IRM occurs in fields of less than 150 mT, and the saturation IRM is usually more resistant to AF demagnetization than the NRM (Fig. 6E). This behavior is typical of samples in which the remanence is dominated by multidomain magnetite, and this phase likely carries the pronounced VRM in these samples. The streaked distribution toward northwest declinations and shallow inclinations may be due to the presence of small amounts of fine-grained, single domain or pseudosingle domain magnetite which carries the B_1 magnetization. The magnetite probably occurs as fine inclusions in plagioclase and/or ferromagnesian minerals. Two thermomagnetic curves from magnetic extracts showed reversible behavior and Curie temperatures of 560 and 565 °C indicating they are low-Ti titanomagnetite.

Estimating the grand-mean direction for the group B dikes is difficult owing to the large number of samples that did not yield stable endpoints. Given the large number of samples that only yielded great circles, most site-mean directions were calculated by calculating remagnetization circles and combining these with stable endpoints to yield estimates of the site-mean directions (McFadden and McElhinny, 1988). Remagnetization circles used in the analysis were well determined, as evidenced by MAD values which ranged from 1.5° to 12°, with 90% less than 5°. Most sites have at least some samples that yielded stable endpoints, and site-mean directions determined by this method yielded well-defined means of northwest declination and shallow negative inclination. Individual site means yielded k values ranging from 15 to 206, with α_{95} values of 4.5° to 23.0° (Fig. 5C; Table 2). The 23 individual sites from the group B dikes are combined to yield what we interpret to be 12 independent readings of the geomagnetic field, with a grand-mean direction of $D = 301.5°$, $I = -17.1°$ ($k = 117$, $\alpha_{95} = 4.0°$) (Fig. 5D, Table 3), which we consider to be the best estimate of the B_1 magnetization.

To assess whether, the B_1 magnetization is primary, five samples from host gneiss from one locality were collected for a baked contact test. All samples were collected within distance of 0.2 m from a 35-m wide group B dike (Fig. 8C) which has a characteristic dike-mean magnetization of $D = 296.6°$ and $I = -11.9°$ ($k = 142$, $\alpha_{95} = 6.4°$, $N = 6$ sites; dike 16, Table 2). These samples were used for isotopic dating, and thus were only demagnetized by AF methods. The gneiss samples contain two magnetization components. The first is of low coercivity and gives a direction of north declination and moderate positive inclination; it is probably a VRM acquired in the present-day field. The second component yields characteristic directions of west declination and shallow negative inclination (Fig. 11A and B), with a site-mean of $D = 269.1°$ and $I = -23.2°$ ($k = 116.5$, $\alpha_{95} = 7.1°$, $n = 5$ samples) (Fig. 8D). Although similar in inclination to that observed from the adjacent dike, the dike and contact zone site-mean directions are distinct at the 95% confidence level (Fig. 8D). This result suggests that the two magnetizations are different and that the baked contact test is negative, which is surprising given the proximity of the gneiss samples to the dike contact and the thickness of the dike. Biotite separated from one of the paleomagnetic baked contact samples in the gneiss at site TR50 gives an apparent age similar to the age of dike emplacement obtained by U–Pb and $^{40}Ar/^{39}Ar$ systems (discussed below) indicating that the gneiss sample was heated to temperatures during dike intrusion sufficient to reset its biotite $^{40}Ar/^{39}Ar$ system. Because of the strongly anisotropic fabric of the host gneiss, the discordance of the dike and gneiss site-mean directions may be due to deflection of the remanence vector by the metamorphic rock fabric. The rock magnetic fabric was not measured prior to destruction of the rock samples for mineral separations for dating purposes, so this hypothesis could not be tested. The results of the paleomagnetic baked contact test alone cannot be used to establish whether the observed B dike magnetization is a primary TRM or is due to a younger remagnetization, but the concordance of the $^{40}Ar/^{39}Ar$ age spectra with hornblendes from the dike and the biotite from the host gneiss sample at this site indicate that the argon systems of these rocks have not been significantly thermally reset since dike emplacement at ca. 780 Ma.

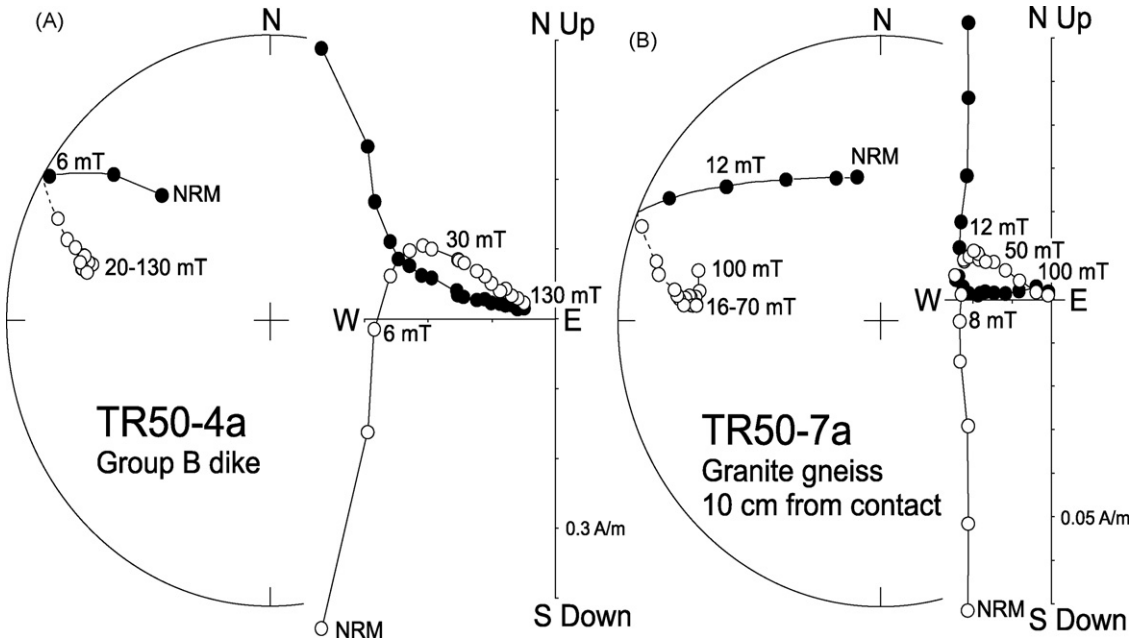


Fig. 11. Paleomagnetic baked contact results from a group B dike. (A) Orthogonal vector diagram and equal area projection showing demagnetization results from dike sample displaying a well-defined characteristic direction of northwest declination and shallow negative inclination. (B) Orthogonal vector diagram and equal area projection of a granite gneiss sample adjacent in the contact zone adjacent to the dike. Gneiss samples from this contact test show a well-resolved characteristic direction that is similar to that of the dike. Both diagrams are in situ coordinates.

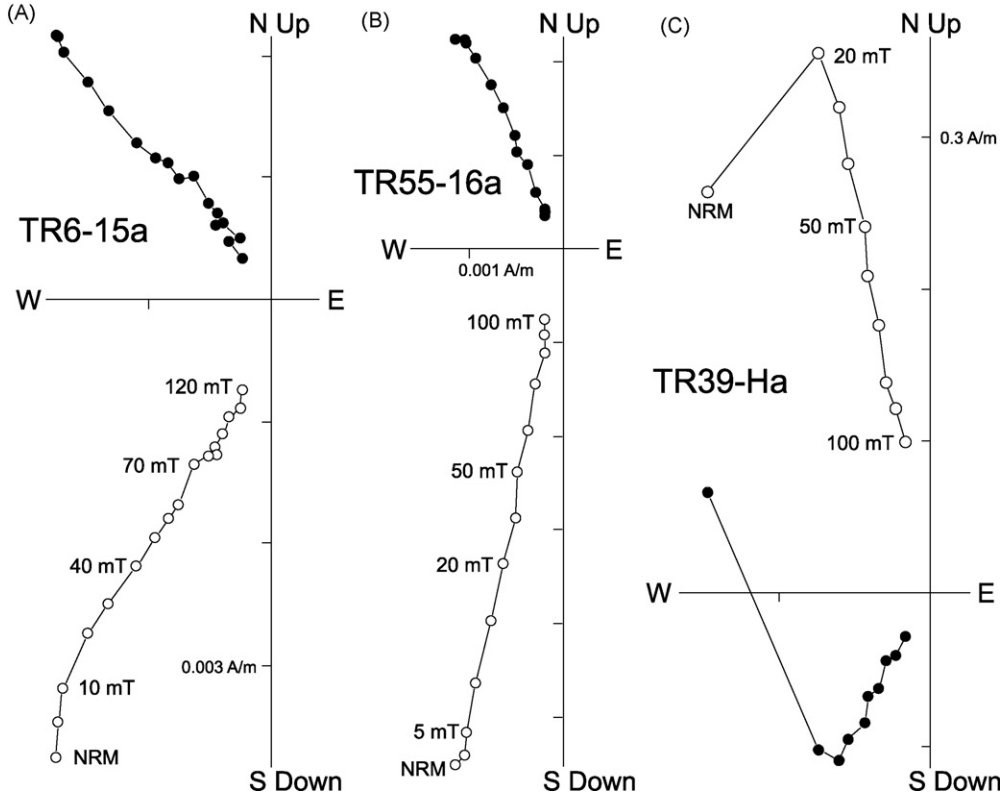


Fig. 12. In situ orthogonal vector diagrams showing magnetizations thought to be Late Cretaceous/Tertiary in age. (A and B) Well-defined magnetizations of northwest declination and moderate inclination obtained from Proterozoic dike samples. (C) Magnetization of southwest declination and moderate negative inclination obtained from a small Cretaceous intrusion in the southern Tobacco Root Mountains. Conventions as in Fig. 4.

4.3. R magnetization

The *R* magnetization is characterized by well-defined, univectorial magnetizations of north to northwest declination and moderate to steep inclination (Fig. 12A and B). In general, this magnetization is mostly restricted to group B dikes, or is found in dikes of uncertain affiliation such as in sites TR55 and 56, from the Mammoth dike in the northern Tobacco Root Mountains (Fig. 2b). Samples that contain the *R* magnetization are characterized by moderate to high MDFs (25–60 mT), distributed laboratory unblocking temperatures below 580 °C, and 95–98% saturation of IRM by 300 mT. These observations indicate that the magnetization is dominated by a low coercivity phase, such as magnetite. A single thermomagnetic analysis of a magnetic extract from a Mammoth dike sample (not shown) gave reversible heating and cooling curves with a Curie temperature of 575 °C, which indicates that the extract is nearly pure magnetite. Petrographic observations indicate that magnetite and ilmenite, as oxidation-exsolution products of titanomagnetite, are common in these rocks. The paleomagnetic and rock magnetic evidence indicates that the *R* magnetization in these samples is carried by low-Ti magnetite.

The northwest declination and positive inclination direction for the *R* magnetization suggests that is of secondary origin and significantly younger than the A_1 or B_1 magnetizations, which we consider primary TRMs. However, its origin is not clear for all samples. In some cases, such as in group B sites that show partial to complete recovery of the B_1 remanence, the *R* magnetization may result from failure to remove the pronounced VRM present in these samples. This appears unlikely, given that most samples with this magnetization have declinations that are significantly west of the present-day or time-averaged field directions. In other cases, the *R* magnetization is associated with highly altered or heavily fractured parts of dikes. At site TR20, for example, the *R* magnetization is in samples from the intensely epidotized dike margin of a group B dike, whereas it is only found in one sample from the fresh dike interior. Similarly, paleomagnetic sites TR55 and TR56 are in a dike pervasively fractured due to Laramide thrust movement along the adjacent Mammoth fault. For such sites, we suspect that the *R* magnetization reflects complete remagnetization by Late Cretaceous or early Tertiary hydrothermal fluids, resulting in a magnetization carried by magnetite.

The NRM directions of the small Laramide pluton are south directed and have steep to moderate negative inclinations, with NRM intensities between 1.55×10^{-1} and 3.03×10^{-1} A/m. AF demagnetization is characterized by moderate to high median destructive fields (50–60 mT) and univectorial decay of the magnetic vector (Fig. 12C). Thermal demagnetization showed unblocking temperatures that ranged from 250 to >620 °C, indicating that this magnetization is carried by a combination of magnetite and hematite. The mean direction is $D = 191.0°$, $I = -68.1°$ ($k = 110$, $\alpha_{95} = 5.8°$) (Fig. 13; Table 2).

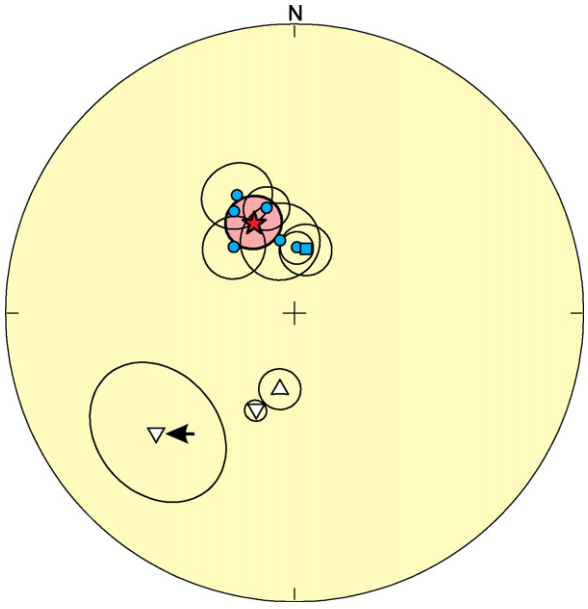


Fig. 13. In situ equal-area projection showing mean directions and α_{95} cones of confidence for dikes that contain the *R* magnetization. The *R* magnetization is considered to be a Late Cretaceous or early Tertiary magnetization that has completely overprinted the remanence in some dike samples. The group-mean direction for the *R* magnetization is shown by the star. Also shown is the site mean direction (square) for a northwest-striking Proterozoic mafic dike in the Madison Range that was remagnetized during Late Cretaceous/early Tertiary time. The upward-pointing triangle shows the site-mean from a Late Cretaceous intrusion exposed in the southern Tobacco Root Mountains; the downward-pointing triangles are paleomagnetic sites from the Tobacco Root Batholith reported by Hanna (1973); the site-mean shown by the arrow was not used in calculating a summary Late Cretceous/early Tertiary direction as described in the text because of its large α_{95}.

5. Discussion

Proterozoic mafic dikes in the Tobacco Root of southwest Montana yield three remanent magnetizations. The A_1 and B_1 magnetizations are distinctly different from each other and both are unlike magnetizations associated with any part of the Phanerozoic APW path for North America. As discussed below, we consider them to be primary magnetizations of Proterozoic age. The *R* magnetization is probably of Late Cretaceous or early Tertiary age. To compare the magnetizations from these rocks with reference poles derived from elsewhere in North America, we converted site-mean directions to a common polarity and calculated virtual geomagnetic poles (VGPs) for the three groups, assuming an axially geocentric dipole field.

The A_1 magnetization is found in Wooden et al.'s (1978) geochemical group A dikes and one group C dike (i.e., dike 9, Table 2). The remanence is well defined by dual polarity magnetizations and there is no apparent difference between the paleomagnetic data obtained from the Group A and Group C dikes. Although not antipodal, we consider the best estimate of the A_1 direction to include both the normal and reverse polarity data sets and this yields an in situ grand-mean direction of $D = 225.0°$, $I = 61.8°$ ($k = 27.9$, $\alpha_{95} = 7.7°$, $N = 14$ independent readings of the geomagnetic field). Averaging the VGPs gives a paleomagnetic pole located in the central Pacific Ocean at

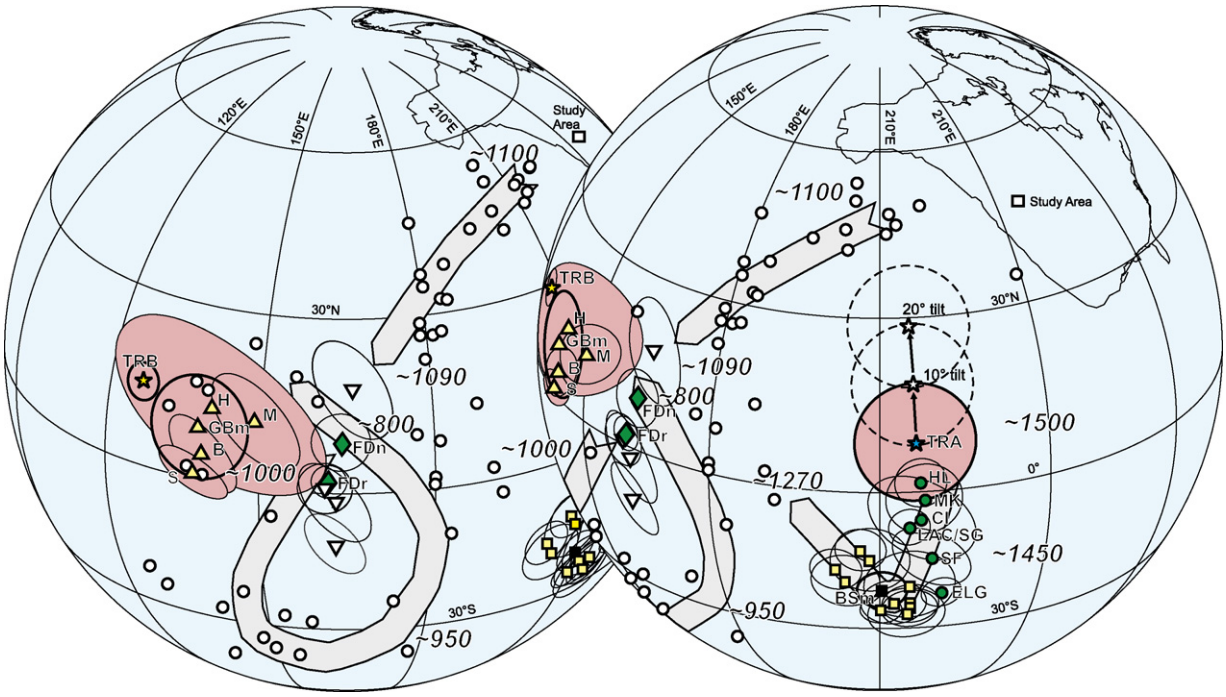


Fig. 14. Apparent polar wander path for North America from ca. 1500 to 700 Ma showing the Meso- and Neoproterozoic paleomagnetic poles from the Tobacco Root Mountains geochemical group A and B dikes. Stars are mean poles from this study, filled circles are poles from ca. 1450 Ma igneous intrusions, squares are poles from the Mesoproterozoic Belt Supergroup, open circles are Meso- and Neoproterozoic poles, downward-pointing triangles are poles from sedimentary strata of the Uinta Mountains and Grand Canyon, upward-pointing triangles are poles from ca. 780 Ma intrusions of the Gunbarrel magmatic event, diamonds are poles from the 723 Ma Franklin Dikes. Also shown are 95% confidence limits for selected poles. TRA = mean pole for the Tobacco Root Mountains A_1 magnetization Arrow and open circles show the movement of the pole due to potential tilt corrections as described in the text. HL = Harp Lake; MK = Michikamau intrusion; CI = Croker Island; LAC/SG = Laramie anorthosite complex/Sherman Granite; SF = St. Francois Mountains Volcanics; ELG = Electra Lake Gabbro; BSm = mean of Belt Supergroup Poles; TRB = mean pole for the Tobacco Root Mountains B_1 magnetization; H = Hottah mafic sheets; S = McKenzie dikes; B = Christmas Lake dike from the Beartooth Mountains, Montana; M = mafic dike at Mount Moran, Wyoming; GBm = mean of Gunbarrel dike poles; FDn and FDr and normal and reverse polarity Franklin Dikes, respectively. Sources of data: Elston et al. (2002), Harlan et al. (1994, 1997), Harlan and Geissman (1998), Meert and Stuckey (2002), Park and Jefferson (1991), Park et al. (1995a), Weil et al. (2003), Weil et al. (1998, 2004, 2006), and Warnock et al., 2000).

216.1°E, 8.7°N ($K = 15.8$, $A_{95} = 10.3°$) (Table 3). Remanence acquisition in the A_1 dikes spanned at least two polarity chrons, or a minimum of several tens of thousands of years indicating that the A_1 magnetization has averaged some paleosecular variation (PSV). The fact that the normal and reverse polarity directions are not antipodal may indicate that PSV has not been fully averaged or that the age of emplacement of the normal and reverse polarity dikes and dikes of group A and C resulted from discrete but temporally restricted magmatic events during the Mesoproterozoic. However, the precision of available geochronology does not allow such a difference in age to be distinguished.

The presence of essentially identical magnetizations in dikes of geochemical groups A and C, which provided Rb–Sr dates of 1455 and 1100 Ma, respectively, suggests that the geologic reliability of at least one of the Rb–Sr dates is questionable given the similarity in paleomagnetic directions from the group A and C dikes. Comparison of the combined group A and C poles, which plot at near equatorial latitudes and ca. 1100 Ma poles that plot at significantly higher north latitudes and more western longitudes (Fig. 14) indicate that the 1100 Ma Rb–Sr date is not supported by the paleomagnetic results from this study. Sm–Nd dating of minerals from a paleomagnetic core sample of a reverse polarity Group A dike yields an age of 1448 ± 49 Ma (Harlan et al., 2005). This age confirms the validity of the group A Rb–Sr age of Wooden et al. (1978), but is more precise. This age is considered to be the best estimate for the emplacement age of the group A dikes. The similarity between paleomagnetic directions between the group A and C dikes suggests that they were probably emplaced at about the same time. Furthermore, as discussed below, the paleomagnetic poles from the group A and C dikes are similar to poles from ca. 1450 Ma rocks elsewhere in North America (e.g. Harlan and Geissman, 1998) and are significantly different from poles of ca. 1100 Ma. Thus, we suggest that the 1100 Ma Rb–Sr date reported for the group C dikes is not reliable.

Assuming that the A_1 magnetization is a primary TRM and that the Wyoming and Laurentian cratons were joined by Mesoproterozoic time (Hoffman, 1988; Dutch and Nielsen, 1990; Mueller and Frost, 2006; Foster et al., 2006), we compare the A_1 paleomagnetic pole with poles from rocks of similar age elsewhere in North America (Fig. 14). The in situ A_1 pole plots slightly north of the main locus of poles from ca. 1450 to 1420 Ma intrusions from Labrador, the central U.S., and the Rocky Mountains. The apparent agreement between the A_1 pole and the other 1450–1420 Ma poles is remarkable given that the A_1 pole has not been corrected for potential post-emplacement tilting.

The group B dikes give an in situ grand-mean direction of $D = 301.5°$, $I = -17.1°$, $k = 117$, $\alpha_{95} = 4.0°$, $N = 12$ independent readings), with a paleomagnetic pole in the western Pacific Ocean at 14.6°N, 127.0°E, ($K = 190$, $A_{95} = 3.2°$) (Table 3), north of present-day New Guinea (Fig. 14). U–Pb and $^{40}Ar/^{39}Ar$ geochronologic data from group B dikes reported in Harlan et al. (2005) and this study was obtained from hand samples collected from the same rocks sampled at paleomagnetic sites TR6, TR19, TR21, TR62, and TR50; these data indicate that these dikes were intruded at 780 Ma. Furthermore, the VGP from the group B dikes is concordant with VGPs from Neoproterozoic mafic intrusions elsewhere in western North America, including the northwest-striking Christmas Lake and Mount Moran dikes in the nearby Beartooth and Teton Ranges, respectively (Harlan et al., 1997), the Hottah sheets of the northwestern Canadian Shield (Park et al., 1995a) and mafic dikes in the Mackenzie Mountains of the Northwest Territories (Park, 1981; Park et al., 1989) (Fig. 14). Several of these rocks (the Hottah sheets, Mackenzie dikes, and the Christmas Lake dike) give isotopic dates of ca. 780 Ma by either the U–Pb or $^{40}Ar/^{39}Ar$ methods (Heaman et al., 1991; LeCheminant and Heaman, 1994; Park et al., 1995b; Harlan et al., 2003b). The VGP is also similar to the normal and reverse polarity poles from the 723 Ma Franklin dikes of northern Canada (Park and Jefferson, 1991; Heaman et al., 1991) (Fig. 14) and some sedimentary rocks from the Uinta Mountains Supergroup and the Galeros Formation of the Chuar Group of the Grand Canyon Supergroup (Weil et al., 2004, 2006).

The concordance of both the group A_1 and B_1 poles with those from rocks of similar age is striking and implies that the basement rocks into which they intrude have not been significantly tilted since emplacement of the dikes at ca. 1450 and 780 Ma. However, small-magnitude post-emplacement tilting cannot be unequivocally ruled out because of the angular discordance that does exist between the Tobacco Root group A_1 dike pole and those from the Belt Supergroup that plot significantly to the south (Fig. 14). The rocks from which the Belt poles are derived were emplaced between 1470 and 1400 Ma (Evans et al., 2000), with the poles from the Belt Supergroup interpreted to represent a stillstand in the North American APW during this time (Elston et al., 2002). Numerous workers, as early as the late 1970's (e.g. Irving, 1979), have noted that Belt Supergroup poles plot significantly south of most ca. 1400 Ma poles from elsewhere in North America, which are mostly from intrusive rocks that lack a record of paleohorizontal. At least some of this discordance between poles derived from these igneous rocks broadly distributed across Laurentia and those of the Belt Supergroup could be the result of post-emplacement tilting of the intrusive rocks, as well as failure of some of the poles from intrusive rocks, such as the Tobacco Root dikes, to fully average paleosecular variation. Nonetheless, neither of these explanations adequately explain why most poles from intrusions are discordant with those of the Belt Supergroup. An additional possibility is that the areas where most of the Belt Supergroup data have been obtained have experienced some degree of vertical axis rotation since remanence acquisition, as explored by Elston et al. (2002). Furthermore, despite magnetic and field-based tests (e.g. reversal and fold tests) of the antiquity of the remanence in the Belt Supergroup rocks, it remains possible that the rocks were remagnetized over along period of time prior to deformation and that workers have been comparing magnetizations of substantially different age. Another possibility, consistent with the relative distribution of the poles is that the Belt Supergroup poles have been systematically affected by inclination shallowing. At present the discordance between the poles from the ca. 1470 and 1400 Ma sedimentary rocks of the Belt Supergroup and intrusive rocks from Laurentia remains poorly understood.

The extent to which post-emplacement tilting may affect the remanence in intrusions, such as those in this study, is a problem that is endemic to the use of plutonic rocks to determine paleomagnetic poles for tectonic studies. To assess whether post-emplacement tilting may have affected the magnetizations characteristic of group A and B dikes, we compare the paleomagnetic direction from the secondary *R* magnetization with expected directions from reference directions for North America, and compare our observed primary poles with those of equivalent age from elsewhere in North America. These comparisons suggest that post-dike emplacement tilting of the Tobacco Root has been minimal.

The overall mean of the *R* magnetization, which we interpret to be a Late Cretaceous/early Tertiary magnetization, as well as two sites in Late Cretaceous intrusions, after inversion to normal polarity, yield a combined mean of $D = 346.8°$, $I = 64.2°$ ($k = 46.6$, $\alpha_{95} = 8.2°$; Fig. 13, Table 3). The origin of this secondary magnetization is not known with certainty, but we consider it to likely have resulted from remagnetization by hydrothermal fluids associated with Late Cretaceous or Tertiary thermal activity. Evidence for such a remagnetization is compelling along the Mammoth fault in the northern Tobacco Root Mountains where a Laramide thrust fault intensely sheared and fractured the Mammoth diabase dike (Fig. 2B). The fault controlled the emplacement of the Tobacco Root Batholith (Schmidt et al., 1990) and acted as a conduit for late stage magmatic fluids or fluids associated with subsequent precious mineralization. Evidence for fluid activity includes extensive chloritization and epidotization of the dike, and locally intense silicification. The remanence may have been caused by growth of secondary magnetite around original titanomagnetite grains, by growth of new magnetite by precipitation of iron during formation of secondary alteration phases, or by a combination of both processes. Some magnetite may also have formed during oxidation of magmatic sulfides present within the dike. It is unlikely to be the result of widespread tectonothermal activity with temperatures greater than 300 °C given the preservation of Proterozoic ages from biotites in basement rocks from the southern Tobacco Root Mountains, including the biotite from the baked contact zone at site TR50 described above. A similar relationship between intense shearing and fracturing of mafic dikes near or along Laramide faults, hydrothermal fluids, and a magnetite-dominated remagnetization of Proterozoic dikes has been observed in the nearby Highland Mountains and Madison Range (Harlan et al., 1996; Harlan, unpublished data). The age of this secondary magnetization may be closely approximated by the 7 Ma age of the Tobacco Root Batholith (Vitaliano et

al., 1980). Alternatively, it may be somewhat younger as suggested by $^{40}Ar/^{39}Ar$ dates of 67–66 Ma from sericite formed during potassic alteration by hydrothermal fluids in the Pony Mining district located in the northeastern part of the Tobacco Root Mountains (Myers and Snee, unpublished data). The *R* magnetization of dikes in the southern Tobacco Root Mountains probably originated in a similar manner, but complete remagnetization appears to have been limited to some samples from dike margins or from zones of pervasive Laramide fractures that cut the basement rocks and diabase dikes. Most dikes in the southern Tobacco Root Mountains are not found near major Laramide faults or shear zones and thus escaped extensive remagnetization.

Comparison of the *R* and Late Cretaceous/early Tertiary magnetization (Table 3) with the expected 75 Ma reference direction ($D = 352.6^\circ$, $I = 66.7^\circ$, $\alpha_{95} = 3.1^\circ$) for the study area based on the pole from the Adel Mountain Volcanics of western Montana (Gunderson and Sheriff, 1991) yields a statistically indistinguishable difference of $2.6 \pm 6.4^\circ$ using the method of Debiche and Watson (1995). Assuming that the *R* magnetization is Late Cretaceous or early Tertiary in age, any appreciable tilting of the basement rocks must have occurred prior to this time. Furthermore, the presence of flat-lying 53–33 Ma volcanic rocks north of Virginia City (Marvin et al., 1974) indicate that post-middle Tertiary tilting, possibly associated with late Cenozoic extension is minimal.

Because emplacement of the Late Cretaceous Tobacco Root Batholith was synchronous with displacement along northwest-striking, Laramide-age basement-involved faults in the Tobacco Root Mountains (Schmidt et al., 1990), it is possible that at least some tilting of the range may have occurred prior to intrusion of the batholith. Overall, the Tobacco Root Mountains have been proposed to be tilted to the east or northeast, although a specific magnitude for tilting has not been quantified (Burger et al., 2004; Harms et al., 2004). To investigate the effects of possible northeast-tilt of the basement rocks on the paleomagnetic data reported here we have corrected the paleomagnetic poles for tilt about a northwest-striking axis of N50°W, the overall strike of major faults that cut the core of the range. We correct the poles from the group A and B dikes for 10° and 20° northeast-side-down tilt to show how this would affect their position with respect to age-equivalent poles. Greater magnitude of tilt is not justified given that the Proterozoic dikes in the Tobacco Root Mountains are near vertical. Application of this tilt correction increases the discordance between the mean pole from the group A dikes and poles from the ca. 1450 Ma intrusions and Belt Supergroup rocks (Fig. 14). In contrast, the model tilt correction has little or no affect on the pole from the group B dikes (tilt correction results not shown) because the strike of the tilt axis is sub-parallel to that of the B_1 declination and the mean inclination is shallow. For example, correcting the pole from the group B dikes for 10° and 20° tilts results in an angular difference between the in situ and tilt corrected poles of $1.9 \pm 3.4^\circ$ and $4.8 \pm 3.4^\circ$, respectively.

The discrepancy between the Tobacco Root group A_1 pole and those from the Belt Supergroup is problematic because the dikes and the sedimentary rocks are identical in age, although the age of remanence in the Belt Supergroup rocks remains open to question. Furthermore, the group A dikes are considered to have been emplaced during extension accompanying development of the Belt Basin. Comparison of the group A_1 pole with the mean Belt Supergroup pole (Fig. 14) yields an angular discordance of $29.1 \pm 8.1^\circ$. Assuming that the group A dikes adequately averaged paleosecular variation and that the Belt Supergroup remanence is of equivalent age, this discordance would require a tilt correction of 58° (west-side-down) about a strike of N6°E to bring the group A dike pole into coincidence with poles of the Belt Supergroup. Unless this tilting occurred prior to the emplacement of the younger group B dikes, this would displace the group B pole to 57.0°N, 162.3°E, making it significantly discordant with respect to poles from the other 780 Ma intrusions (e.g. Gunbarrel poles). There appears to be no evidence that a west-side-down tilt of such magnitude affected the basement rocks of southwestern Montana.

Overall, the general agreement between the ca. 1450 Ma pole from the group A dikes with other ca. 1450 Ma intrusions, as well as the excellent agreement between the pole from the group B dikes with those from other 780 Ma dikes (Fig. 14), indicates that the basement rocks of the Tobacco Root Mountains have not experienced significant tilting or vertical-axis rotation since emplacement of the dikes in the Meso- and Neoproterozoic time. Although some of the discordance between the pole from the group A and C dikes and those of the Belt Supergroup could reflect incomplete averaging of paleosecular variation, the overall disagreement between ca. 1.45 to 1.4 Ga poles derived from most igneous rocks from North America and those from the Belt Supergroup continues to be problematic.

6. Conclusions

Proterozoic mafic dikes from the southern Tobacco Root Mountains contain three distinct magnetizations that provide insight and improved understanding of the geologic, magmatic, and tectonic evolution of southwestern Montana and the western Cordillera. In addition, they provide key paleomagnetic poles that contribute to increased resolution of the Meso- and Neoproterozoic parts of the North American APW path.

Paleomagnetic directions from geochemically defined group A and C dikes yield similar magnetizations indicating that remanence acquisition in both groups occurred at about the same time. Concordant Rb–Sr and Sm–Nd dates of 1455 and 1448 Ma indicate that the dikes were emplaced during the Mesoproterozoic; baked contact tests indicate that the observed remanent magnetization is a thermoremanent magnetization acquired during dike emplacement. A mean paleomagnetic pole from the group A dikes is consistent with those of most ca. 1450 Ma intrusions from elsewhere in North America, but is discordant with age equivalent strata of the nearby Belt Basin. The reason for this discordance is unknown, but is not due to significant tilting of the basement rocks of the Tobacco Root Mountains.

Paleomagnetic results from group B dikes yield a magnetization that is considered to be primary. U–Pb baddeleyite analyses indicate that the dikes were emplaced at 782 Ma, and hornblende $^{40}Ar/^{39}Ar$ dates from the group B dikes are consistent with this

age. Biotite from a contact zone from one of the dikes yields a reset $^{40}Ar/^{39}Ar$ date of 788 Ma. Together, these data indicate that most of dikes have not been significantly thermally disturbed since emplacement in the Neoproterozoic. Overall, the isotopic dates from the Tobacco Root Mountains document two distinct episodes of emplacement of subparallel northwest-striking mafic dikes, separated by an interval of ~700 Ma. The paleomagnetic pole from the group B dikes is consistent with those obtained from other 780 to 770 Ma intrusions elsewhere in the western Cordillera. As noted by Park et al. (1995b), this consistency demonstrates the tectonic integrity of the western Cordillera and indicates that little or no relative rotation has occurred between these regions since 780 Ma. Together, these ca. 780 Ma dikes are part of regional magmatic episode termed the Gunbarrel Mafic Magmatic Event that may have been a precursor to crustal extension and rifting that resulted in the break up of Rodinia and development of the proto-Pacific Ocean (Park et al., 1995b; Harlan et al., 2003b, 2005). A combined and revised paleomagnetic pole utilizing data from this study and other 780 Ma Gunbarrel dikes and sills (Park, 1981; Park et al., 1989; Park et al., 1995a; Harlan et al., 1997) yields a pole at 9.2°N, 138.7°E ($K = 72.6$, $A_{95} = 9.0°$, $N = 6$ VGPs) and provides a key reference pole for Neoproterozoic plate reconstructions, especially if matching dike segments of identical age can be found on now widely dispersed continental fragments.

Although primary magnetizations were recovered from most Proterozoic dikes sampled during this study, some dikes showed evidence of partial to complete remagnetization yielding well-defined magnetizations identical to those expected from latest Cretaceous to early Tertiary time. The origin of this magnetization is not clear, but in some cases is thought to be due to heavy fracturing of dikes adjacent to Laramide-age fault zones and hydrothermal fluid flow causing growth of new magnetite. Remagnetization due to hydrothermal activity due to Laramide intrusive activity and faulting has been proposed to account for apparent Late Cretaceous remagnetization of Proterozoic dikes elsewhere in Montana. A combined mean direction from the dikes that contain the Late Cretceous remagnetization and paleomagnetic data from Late Cretaceous intrusions associated with the Tobacco Root Batholith yield a mean direction identical to that for the expected 75 Ma reference direction for the study area. The combination of data for all age rocks indicates that the Tobacco Root Mountains have not experienced significant tilting or vertical-axis rotation since emplacement in Meso- and Neoproterozoic time.

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